AI On-device 프로젝트

Team STEP

생산 공정 모니터링 시스템

AI 시스템 반도체 SW 개발자 2기

팀원 고지현, 김선호, 박상현, 윤홍은

목 차

01 프로젝트 개요

02 시스템 설계 및 구현

03 핵심 기술

04 시스템 기능 및 통신

05 결론

1. 프로젝트 개요

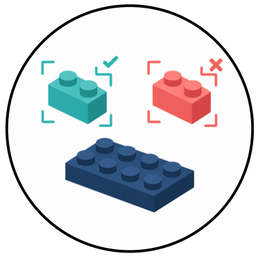
프로젝트 배경

기존 공정의 문제점

프로젝트 목적

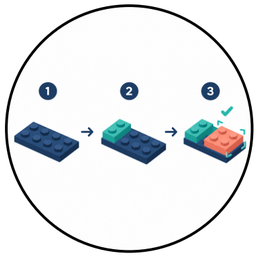
조립오류 단계 별 확인, 작업이력을 연결해 품질의 일관성을 높이고, 재작업과 관리자의 확인 부담을줄이는 것을 목표로 함

프로젝트 목표



AI 기반 조립 불량 검출

부품 누락, 오부품 인식
조립 오류 확인



STEP 별 조립 순서, 위치 검증

단계별 부품 위치,
조립 순서 검사

실시간 작업 현황 모니터링

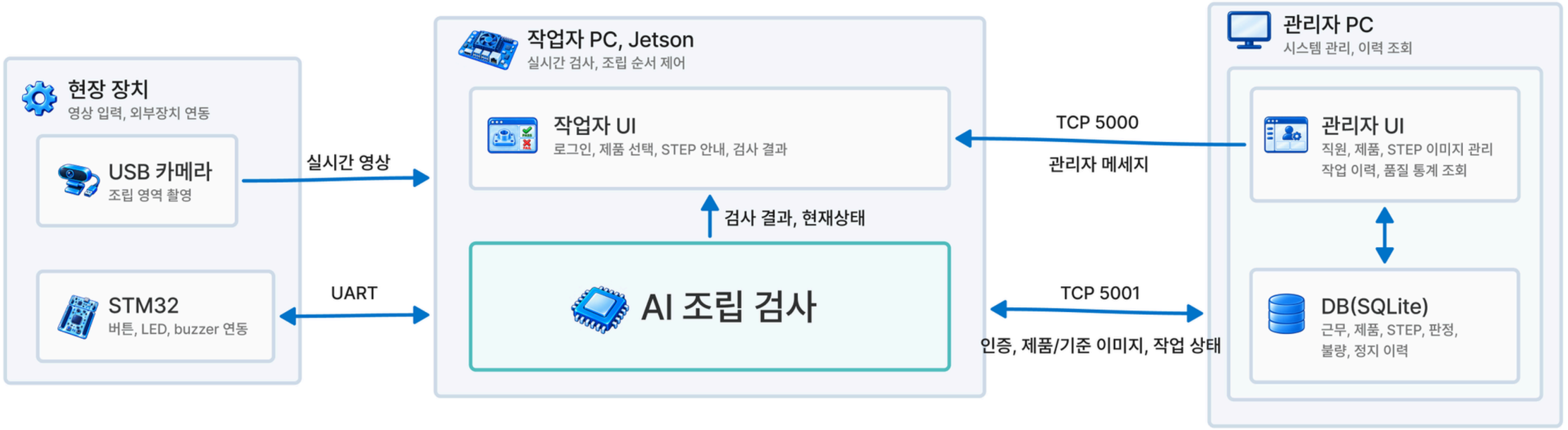
작업자의 진행 상태,
검사 결과 실시간 확인

작업 이력, 품질 데이터 통합 관리

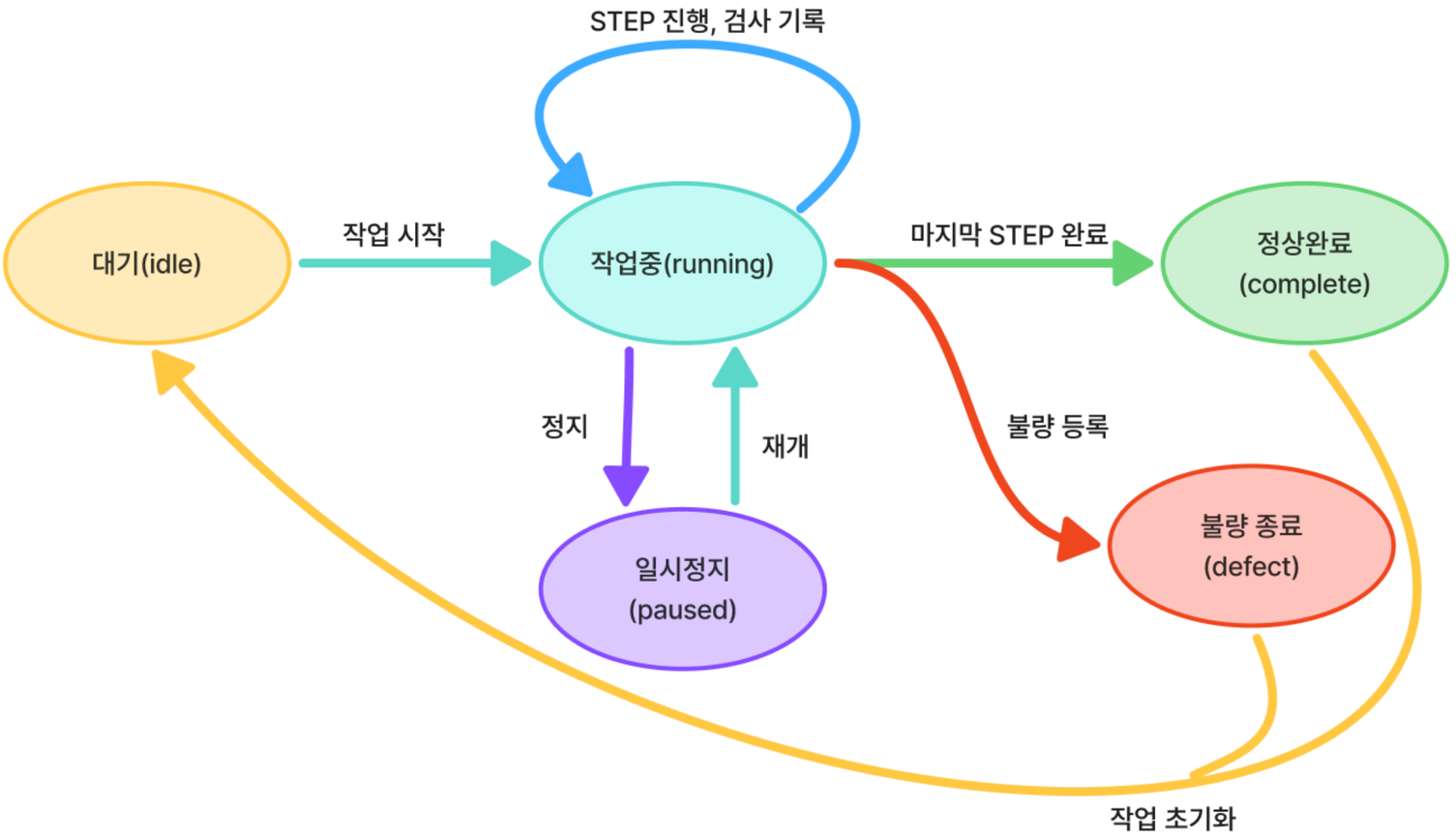
작업 시간, 불량 이력 저장
품질 통계 제공

2. 시스템 설계 및 구현

전체 시스템 흐름


현장 장치
영상 입력, 외부장치 연동
USB 카메라
조립 영역 촬영
STM32
버튼, LED, buzzer 연동
실시간 영상
UART
작업자 PC, Jetson
실시간 검사, 조립 순서 제어
작업자 UI
로그인, 제품 선택, STEP 안내, 검사 결과
검사 결과, 현재상태
AI 조립 검사
TCP 5000
관리자 메세지
TCP 5001
인증, 제품/기준 이미지, 작업 상태
관리자 PC
시스템 관리, 이력 조회
관리자 UI
직원, 제품, STEP 이미지 관리
작업 이력, 품질 통계 조회
DB(SQLite)
근무, 제품, STEP, 판정,
불량, 정지 이력

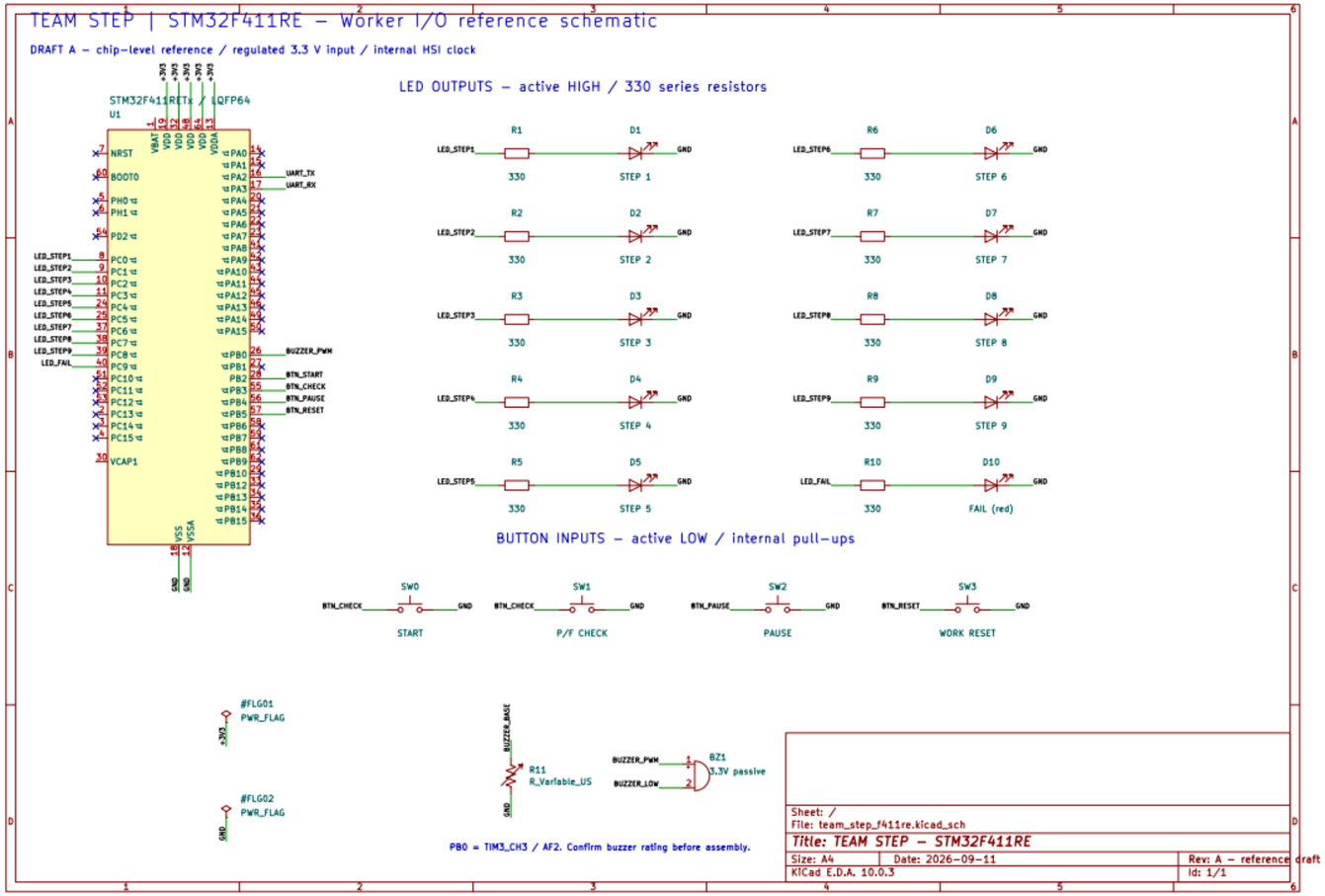
상태천이도


STEP 진행, 검사 기록
대기(idle)
작업 시작
작업중(running)
마지막 STEP 완료
정상완료
(complete)
정지
재개
불량 등록
일시정지
(paused)
불량 종료
(defect)
작업 초기화

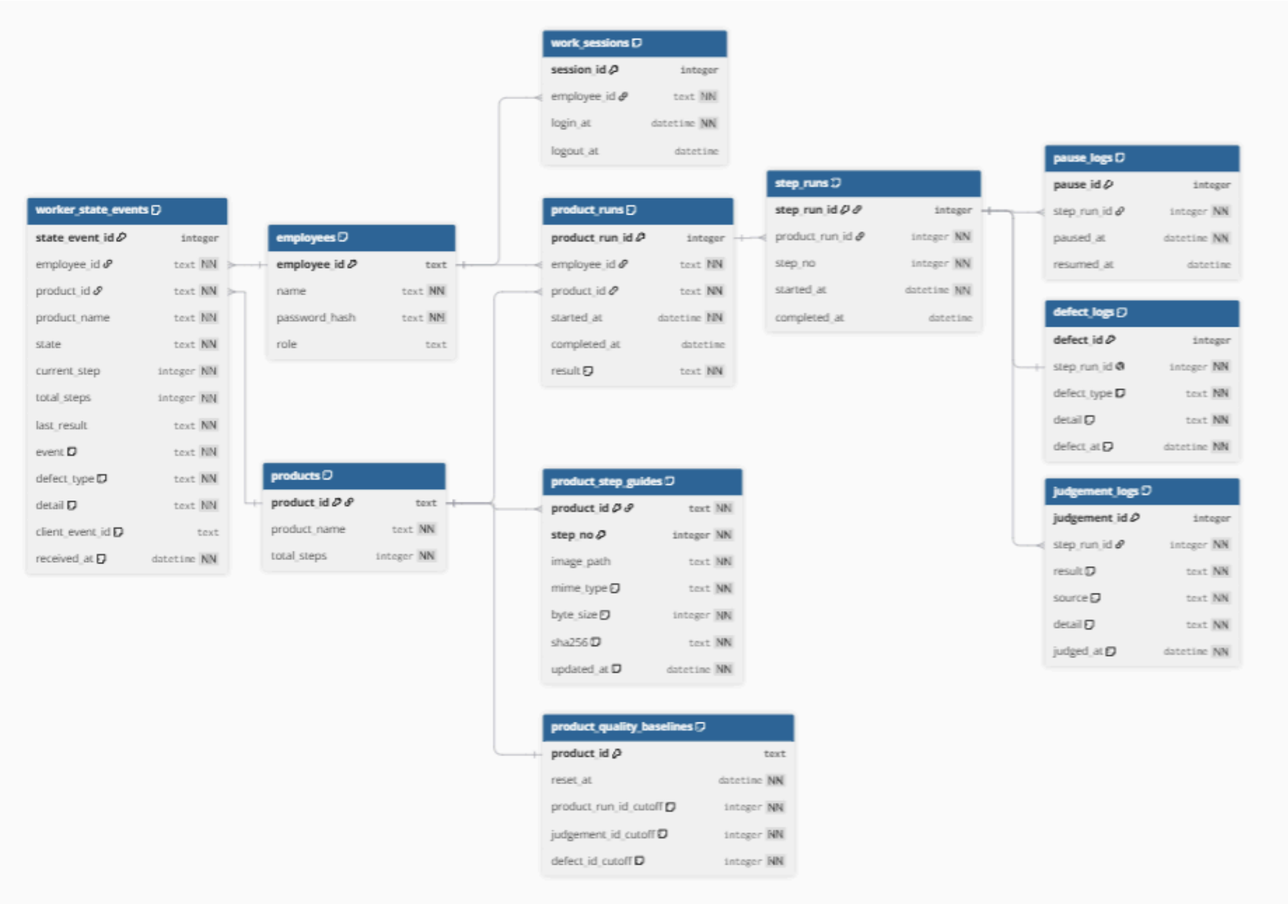
Part List

No	부품	부품 번호	용도	수량
1	STM32	NUCLEO-F411RE	중앙제어 및 QT와 통신 연결	1
2	Jetson Orin Nano	P3766	AI학습을 통한 PASS/FAIL 판단	1
3	웹캠	C270	공정 물품 판단을 위한 카메라	1
4	PC	일체형PC	Worker, Admin 프로그램 실행	2
5	buzzer	SM-1205c	진행 상황 별 부저음	1
6	LED	RED, GREEN	STEP 단계, FAIL 표시	10
7	저항	330Ω	LED 저항	10
8	버튼	택트스위치	ui 제어	3
9	가변 저항		부저 음량 제어	1

회로도


TEAM STEP | STM32F411RE — Worker I/O reference schematic
DRAFT A — chip-level reference / regulated 3.3 V input / internal HSI clock
STM32F411RETx / LQFP64
U1
LED OUTPUTS — active HIGH / 330 series resistors
R1 330 D1 STEP 1
R2 330 D2 STEP 2
R3 330 D3 STEP 3
R4 330 D4 STEP 4
R5 330 D5 STEP 5
R6 330 D6 STEP 6
R7 330 D7 STEP 7
R8 330 D8 STEP 8
R9 330 D9 STEP 9
R10 330 D10 FAIL (red)
BUTTON INPUTS — active LOW / internal pull-ups
SW0 START
SW1 P/F CHECK
SW2 PAUSE
SW3 WORK RESET
R11 R_Variable_US
BZ1 3.3V passive
PB0 = TIM3_CH3 / AF2. Confirm buzzer rating before assembly.
Sheet: /
File: team_step_f411re.kicad_sch
Title: TEAM STEP — STM32F411RE
Size: A4
Date: 2026-09-11
Rev: A — reference draft
KiCad E.D.A. 10.0.3
Id: 1/1

DB Table

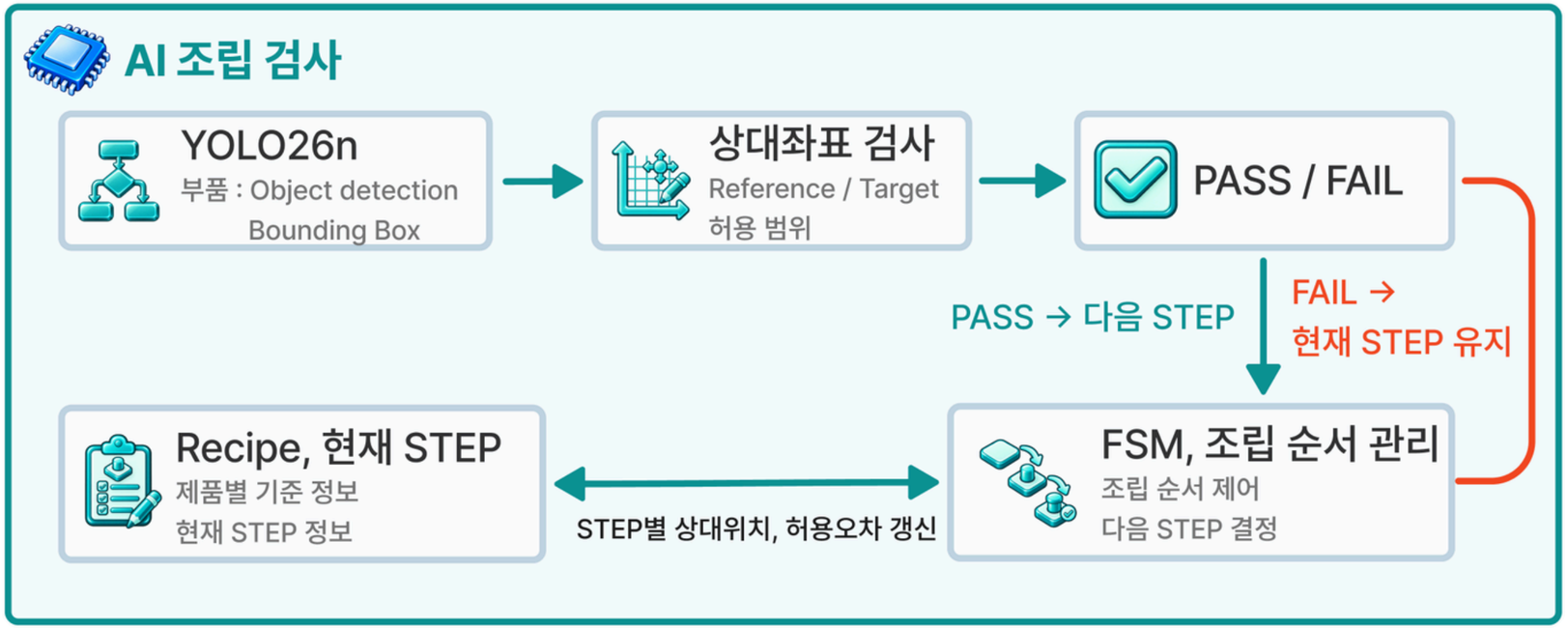

work_sessions
session_id integer
employee_id text NN
login_at datetime NN
logout_at datetime
worker_state_events
state_event_id integer
employee_id text NN
product_id text NN
product_name text NN
state text NN
current_step integer NN
total_steps integer NN
last_result text NN
event text NN
defect_type text NN
detail text NN
client_event_id text
received_at datetime NN
employees
employee_id text
name text NN
password_hash text NN
role text
product_runs
product_run_id integer
employee_id text NN
product_id text NN
started_at datetime NN
completed_at datetime
result text NN
step_runs
step_run_id integer
product_run_id integer NN
step_no integer NN
started_at datetime NN
completed_at datetime
pause_logs
pause_id integer
step_run_id integer NN
paused_at datetime NN
resumed_at datetime
defect_logs
defect_id integer
step_run_id integer NN
defect_type text NN
detail text NN
defect_at datetime NN
judgement_logs
judgement_id integer
step_run_id integer NN
result text NN
source text NN
detail text NN
judged_at datetime NN
products
product_id text
product_name text NN
total_steps integer NN
product_step_guides
product_id text NN
step_no integer NN
image_path text NN
mime_type text NN
byte_size integer NN
sha256 text NN
updated_at datetime NN
product_quality_baselines
product_id text
reset_at datetime NN
product_run_id_cutoff integer NN
judgement_id_cutoff integer NN
defect_id_cutoff integer NN

시연 영상

생산 공정 모니터링 시스템
시연 영상

3. 핵심 기술

AI의 역할


AI 조립 검사
YOLO26n
부품 : Object detection
Bounding Box
상대좌표 검사
Reference / Target
허용 범위
PASS / FAIL
PASS → 다음 STEP
FAIL →
현재 STEP 유지
Recipe, 현재 STEP
제품별 기준 정보
현재 STEP 정보
STEP별 상대위치, 허용오차 갱신
FSM, 조립 순서 관리
조립 순서 제어
다음 STEP 결정

Why YOLO26n?

검사에 필요한 정보는 **“어떤 부품이, 몇개, 어디에 있는가”**이다.

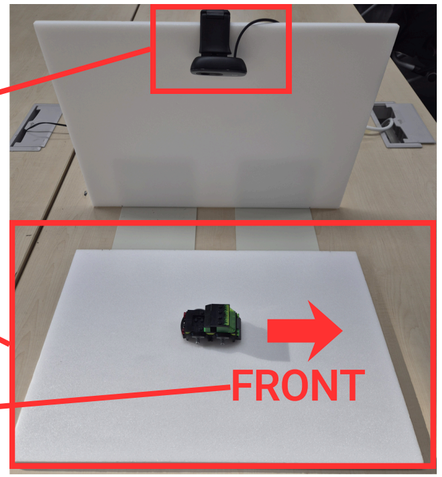
비교항목	Classification	Segementation	Object Detection
제공 정보	이미지 전체의 분류 결과	부품의 픽셀 단위 영역, 윤곽	**부품 종류 + Bounding Box**
제공하는 정보	전체 상태 분류는 가능 개별 부품의 개수·위치는 제공 X	정확한 윤곽·면적을 검사할 때 유용	**부품 개수·상대 위치 검사에 필요한 정보 제공**
데이터 준비	이미지마다 분류 라벨 지정	부품 경계에 맞춘 마스크와 종류 지정	**부품마다 Bounding Box를 그리고 Class 지정**
선택 판단	개별 부품 위치 정보 부족	정밀한 윤곽 정보의 필요성이 낮음	**필요한 정보를 비교적 간단한 라벨링으로 확보**

YOLO26n 선정 이유: 작은 조립 부품의 검출 특화 + Jetson에서의 실행을 고려한 경량화

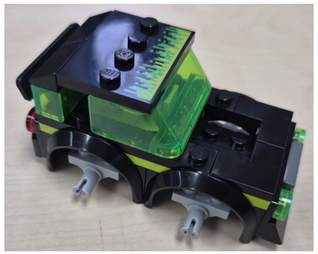
모델 학습 설계

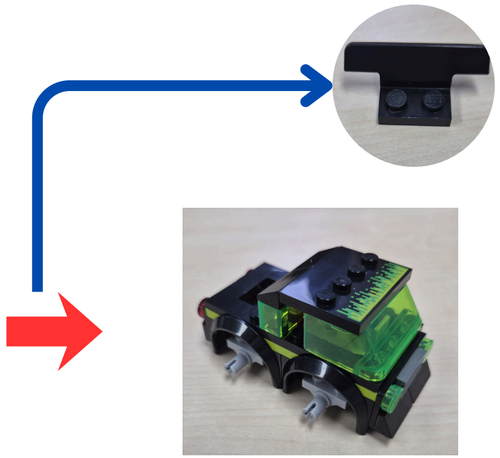
어떻게 하면 **“새 레시피가 추가될 때마다 필요한 촬영/라벨링을 줄일 수 있을까?”**

운용 환경



제품 추가: 기존 부품을 거의 그대로 사용하되 조립 Step만 변경

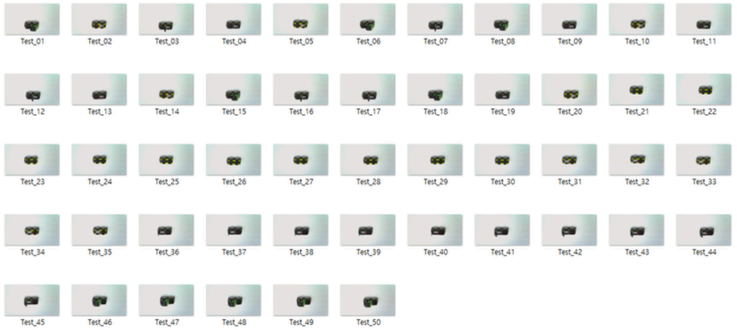




실험 주제: 최소한의 데이터만 학습해도 새로운 제품을 추가했을 때 안정적인 detection이 가능한가? ▶ 모델 1~5 실험 시작

평가 Dataset

Dataset1 - Recipe Test

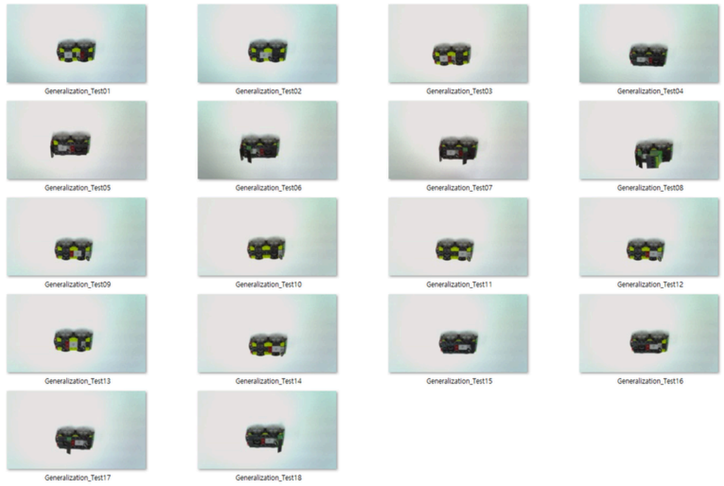


Recipe 1·2의 Step별 이미지 5장씩, 총 50장

Recipe1 - 6 Steps

Recipe2 - add 4 Steps

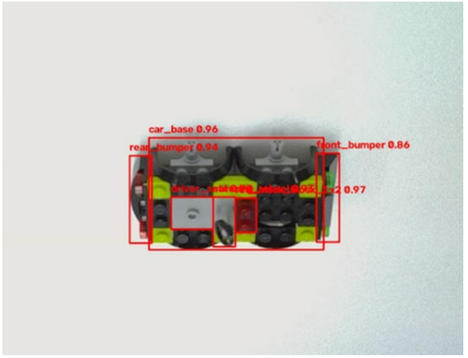
Dataset2 - Generalization Test



상식 선에서 발생할 수 있는 실수를 고려한 부품 배치 18장

평가 지표

지표	의미
Precision(P)	검출한 부품 중 정답의 비율 → **오검출 확인**
Recall(R)	실제 부품 중 찾아낸 비율 → **미검출 확인**
mAP*50	IoU* 0.5 **기준**의 클래스 평균 검출 성능



***mAP(Mean Average Precision)**: 객체 검출 모델이 **여러 종류의 객체**를 얼마나 정확하게 찾는지 나타내는 지표

***IoU** = 예측 박스와 정답 박스가 겹치는 영역 / 두 박스가 차지하는 전체 영역

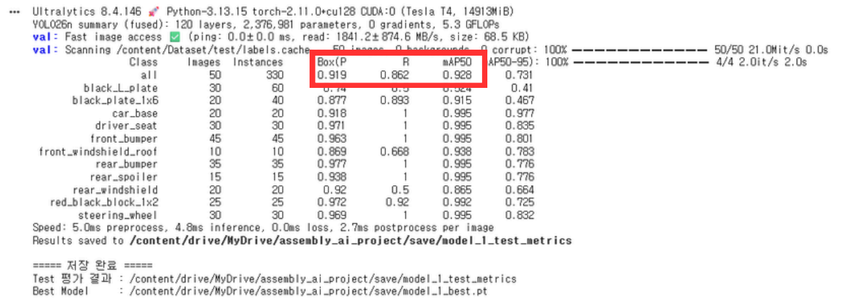
mAP는 **부품을 잘 찾는 지**에 대한 지표(P/F 판단 X) → **P/F 판단은 부품 간 상대 거리 계산 필요**

Model 1 - 하나의 제품으로 시작한 기준 모델

Train 60장 / Validation: 30장, Recipe2 데이터 X

MODEL1			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	91.9	86.2	92.8
Generalization	76.2	83.1	83.0
변화	-15.7	-3.1	-9.8

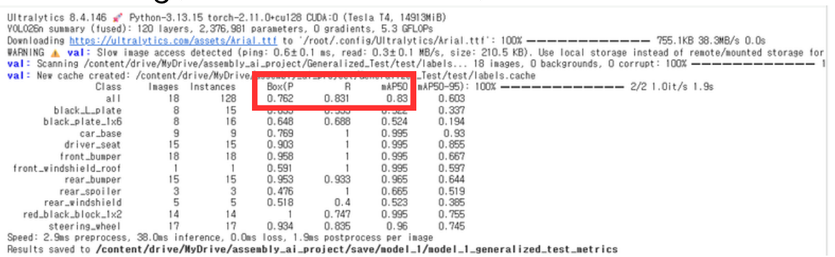
Model 1 Log(Recipe Test)



```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1841.2±874.6 MB/s, size: 68.5 KB)
val: Scanning /content/Dataset/test/labels.cache... 50 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 50/50 21.0Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 4/4 2.0it/s 2.0s
                   all         50        330      0.919      0.862      0.928      0.731
          black_L_plate         30         60      0.914      0.917      0.954       0.41
      black_plate_1x6           20         40      0.877      0.893      0.915      0.467
              car_base          20         20      0.918          1      0.995      0.977
           driver_seat          30         30      0.971          1      0.995      0.835
          front_bumper          45         45      0.963          1      0.995      0.801
  front_windshield_roof         10         10      0.869      0.668      0.938      0.783
           rear_bumper          35         35      0.977          1      0.995      0.776
           rear_spoiler         15         15      0.938          1      0.995      0.776
        rear_windshield         20         20       0.92        0.5      0.865      0.664
    red_black_block_1x2         25         25      0.972       0.92      0.992      0.725
         steering_wheel         30         30      0.969          1      0.995      0.832
Speed: 5.0ms preprocess, 4.8ms inference, 0.0ms loss, 2.7ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_1_test_metrics

===== 저장 완료 =====
Test 평가 결과 : /content/drive/MyDrive/assembly_ai_project/save/model_1_test_metrics
Best Model     : /content/drive/MyDrive/assembly_ai_project/save/model_1_best.pt
```

Model 1 Log(Generalization Test)



```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 38.3MB/s 0.0s
WARNING ⚠ val: Slow image access detected (ping: 0.6±0.1 ms, read: 0.3±0.1 MB/s, size: 210.5 KB). Use local storage instead of remote/mounted storage for
val: Scanning /content/drive/MyDrive/assembly_ai_project/Generalized_Test/test/labels... 18 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 1
val: New cache created: /content/drive/MyDrive/assembly_ai_project/Generalized_Test/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.0it/s 1.9s
                   all         18        128      0.762      0.831       0.83      0.603
          black_L_plate          8         15      0.855      0.933      0.922      0.337
      black_plate_1x6            8         16      0.648      0.688      0.524      0.194
              car_base           9          9      0.769          1      0.995       0.83
           driver_seat          15         15      0.903          1      0.995      0.855
          front_bumper          18         18      0.958          1      0.995      0.667
  front_windshield_roof          1          1      0.591          1      0.995      0.597
           rear_bumper          15         15      0.953      0.933      0.965      0.644
           rear_spoiler          3          3      0.476          1      0.665      0.519
        rear_windshield          5          5      0.518        0.4      0.523      0.395
    red_black_block_1x2         14         14          1      0.747      0.995      0.755
         steering_wheel         17         17      0.934      0.835       0.96      0.745
Speed: 2.9ms preprocess, 38.0ms inference, 0.0ms loss, 1.9ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_1/model_1_generalized_test_metrics
```

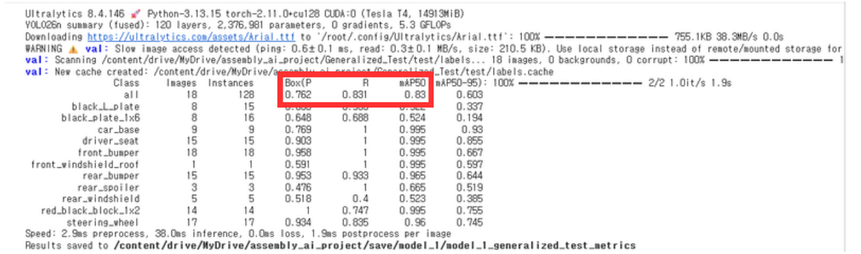
Precision -15.7% → 오검출 증가로 인한 검출 **신뢰도 하락**

Model 2- Single 부품 Image 추가

Train 72장 / Validation: 30장 + Single 부품별 Image 추가

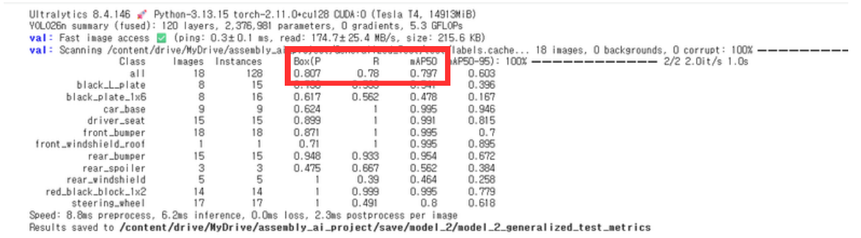
MODEL1			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	91.9	86.2	92.8
Generalization	76.2	83.1	83.0
변화	-15.7	-3.1	-9.8
MODEL2			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	92.5	82.0	91.4
Generalization	80.7	78.0	79.7
변화	-11.8	-4	-11.7

Model 1 Log(Generalization Test)



```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 38.3MB/s 0.0s
WARNING ⚠ val: Slow image access detected (ping: 0.6±0.1 ms, read: 0.3±0.1 MB/s, size: 210.5 KB). Use local storage instead of remote/mounted storage for
val: Scanning /content/drive/MyDrive/assembly_ai_project/Generalized_Test/test/labels... 18 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 1
val: New cache created: /content/drive/MyDrive/assembly_ai_project/Generalized_Test/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 1.0it/s 1.9s
                   all         18        128      0.762      0.831       0.83      0.603
           black_L_plate          8         15      0.888      0.867      0.888      0.337
          black_plate_1x6         8         16      0.648      0.688      0.524      0.194
                 car_base         9          9      0.769          1      0.995       0.93
              driver_seat        15         15      0.903          1      0.995      0.855
             front_bumper        18         18      0.958          1      0.995      0.667
       front_windshield_roof      1          1      0.591          1      0.995      0.597
              rear_bumper        15         15      0.953      0.933      0.965      0.644
              rear_spoiler        3          3      0.476          1      0.665      0.519
             rear_windshield      5          5      0.518        0.4      0.523      0.385
          red_black_block_1x2    14         14          1      0.747      0.995      0.755
             steering_wheel      17         17      0.934      0.835       0.96      0.745
Speed: 2.9ms preprocess, 38.0ms inference, 0.0ms loss, 1.9ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_1/model_1_generalized_test_metrics
```

Model 2 Log(Generalization Test)



```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
val: Fast image access ✅ (ping: 0.3±0.1 ms, read: 174.7±25.4 MB/s, size: 215.6 KB)
val: Scanning /content/drive/MyDrive/assembly_ai_project/Generalized_Test/test/labels.cache... 18 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 2.0it/s 1.0s
                   all         18        128      0.807       0.78      0.797      0.603
           black_L_plate          8         15      0.788      0.688      0.547      0.396
          black_plate_1x6         8         16      0.617      0.562      0.478      0.167
                 car_base         9          9      0.624          1      0.995      0.946
              driver_seat        15         15      0.899          1      0.991      0.815
             front_bumper        18         18      0.871          1      0.995        0.7
       front_windshield_roof      1          1      0.711          1      0.995      0.895
              rear_bumper        15         15      0.948      0.933      0.954      0.672
              rear_spoiler        3          3      0.475      0.667      0.562      0.384
             rear_windshield      5          5          1       0.39      0.464      0.258
          red_black_block_1x2    14         14          1      0.999      0.995      0.779
             steering_wheel      17         17          1      0.491        0.8      0.618
Speed: 8.8ms preprocess, 6.2ms inference, 0.0ms loss, 2.3ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_2/model_2_generalized_test_metrics
```

Precision + 4.5%

Recall - 5.1%, mAP - 3.3%

Model 3- Recipe 2 Step 추가

Train 92장 / Validation: 50장 + Recipe 2에서 변경되는 4 Steps Image 추가

MODEL2			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	92.5	82.0	91.4
Generalization	80.7	78.0	79.7
변화	-11.8	-4	-11.7
MODEL3			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	99.0	99.7	99.5
Generalization	84.8	88.8	90.7
변화	-14.2	-10.9	-8.8

Model 3 Log(Recipe Test)

```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 2012.0±518.1 MB/s, size: 66.9 KB)
val: Scanning /content/Dataset/test/labels... 50 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 50/50 2.0Kit/s 0.0s
val: New cache created: /content/Dataset/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 4/4 1.5it/s 2.7s
                   all         50        330       0.99      0.997      0.995      0.799
         black_L_plate         30         60      0.998          1      0.995      0.792
       black_plate_1x6         20         40          1      0.977      0.995      0.582
              car_base         20         20      0.981          1      0.995      0.938
           driver_seat         30         30      0.989          1      0.995      0.839
          front_bumper         45         45          1      0.988      0.995      0.795
    front_windshield_roof      10         10      0.964          1      0.995      0.939
           rear_bumper         35         35      0.995          1      0.995       0.77
           rear_spoiler        15         15      0.989          1      0.995       0.72
         rear_windshield       20         20      0.989          1      0.995      0.757
    red_black_block_1x2        25         25      0.995          1      0.995      0.826
          steering_wheel       30         30      0.988          1      0.995      0.826
Speed: 5.0ms preprocess, 18.3ms inference, 0.0ms loss, 8.8ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_3_test_metrics

===== 저장 완료 =====
Test 평가 결과 : /content/drive/MyDrive/assembly_ai_project/save/model_3_test_metrics
Best Model     : /content/drive/MyDrive/assembly_ai_project/save/model_3_best.pt
```

Model 3 Log(Generalization Test)

```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
val: Fast image access ✅ (ping: 0.4±0.1 ms, read: 186.8±32.7 MB/s, size: 237.2 KB)
val: Scanning /content/drive/MyDrive/assembly_ai_project/save/model_3/model_3_generalized_test_metrics/labels.cache... 18 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 3.3it/s 0.6s
                   all         18        128      0.848      0.888      0.907      0.654
         black_L_plate          8         15      0.938      0.933      0.944      0.608
       black_plate_1x6          8         16      0.533      0.572      0.501      0.138
              car_base          9          9      0.903          1      0.995      0.928
           driver_seat         15         15      0.935          1      0.995      0.828
          front_bumper         18         18      0.969          1      0.995      0.601
    front_windshield_roof       1          1      0.648          1      0.995      0.697
           rear_bumper         15         15      0.958      0.933      0.955       0.57
           rear_spoiler         3          3      0.475      0.617      0.641      0.488
         rear_windshield        5          5          1      0.718      0.962      0.798
    red_black_block_1x2         14         14      0.999          1      0.995      0.769
          steering_wheel        17         17      0.981          1      0.995      0.771
Speed: 2.7ms preprocess, 11.0ms inference, 0.0ms loss, 1.6ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_3/model_3_generalized_test_metrics
```

mAP(Recipe Test) +8.1 %

mAP(Generalization) +11 %

Model 4- 좌우 반전 증강

Model 3 Data에 Horizontal Flip 적용 (추가 촬영 X)

MODEL3			
평가	Precision(%)	Recall(%)	mAP50-95(%)
Recipe Test	99.0	99.7	79.9
Generalization	84.8	88.8	65.4
변화	-14.2	-10.9	-14.5
MODEL4			
평가	Precision(%)	Recall(%)	mAP50-95(%)
Recipe Test	98.8	99.6	81.5
Generalization	87.8	89.3	69.0
변화	-11.0	-10.3	-12.5

Model 4 Log(Recipe Test)

```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1462.7±589.9 MB/s, size: 66.9 KB)
val: Scanning /content/Dataset/test/labels... 50 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 50/50 1.7Kit/s 0.0s
val: New cache created: /content/Dataset/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 4/4 1.5it/s 2.7s
                   all         50        330      0.988      0.996      0.995      0.815
          black_L_plate         30         60      0.994          1      0.995      0.855
        black_plate_1x6         20         40          1      0.953      0.995      0.721
              car_base          20         20      0.986          1      0.995      0.899
           driver_seat          30         30      0.989          1      0.995      0.848
          front_bumper          45         45      0.996          1      0.995      0.828
  front_windshield_roof         10         10      0.968          1      0.995      0.898
           rear_bumper          35         35      0.993          1      0.995      0.767
          rear_spoiler          15         15       0.98          1      0.995      0.719
         rear_windshield        20         20      0.981          1      0.995      0.814
    red_black_block_1x2         25         25      0.989          1      0.995      0.804
         steering_wheel         30         30      0.991          1      0.995      0.868
Speed: 3.9ms preprocess, 15.9ms inference, 0.0ms loss, 2.2ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_4_test_metrics

===== 저장 완료 =====
Test 평가 결과 : /content/drive/MyDrive/assembly_ai_project/save/model_4_test_metrics
Best Model     : /content/drive/MyDrive/assembly_ai_project/save/model_4_best.pt
```

Model 4 Log(Generalization Test)

```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
val: Fast image access ✅ (ping: 0.4±0.1 ms, read: 211.8±28.9 MB/s, size: 259.3 KB)
val: Scanning /content/drive/MyDrive/assembly_ai_project/Generalized_Test/test/labels.cache... 18 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 2/2 3.4it/s 0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)
                   all         18        128      0.878      0.869      0.903       0.69
          black_L_plate         8         15          1      0.875      0.943      0.804
        black_plate_1x6         8         16      0.703      0.562      0.649      0.217
              car_base          9          9      0.915          1      0.995       0.91
           driver_seat          15         15      0.948          1      0.995      0.855
          front_bumper          18         18       0.97          1      0.995      0.696
  front_windshield_roof         1          1      0.725          1      0.995      0.895
           rear_bumper          15         15      0.957      0.933      0.946      0.673
          rear_spoiler          3          3      0.494      0.654      0.456      0.287
         rear_windshield        5          5      0.981        0.8      0.962      0.838
    red_black_block_1x2         14         14      0.974          1      0.995      0.777
         steering_wheel         17         17      0.991          1      0.995      0.808
Speed: 5.0ms preprocess, 6.4ms inference, 0.0ms loss, 1.3ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_4/model_4_generalized_test_metrics
```

Precision(Generalization) +3.0 %

mAP50-95(Generalization) +4.4 %

Model 5- Single Data 제거

Single Image 제거 및 실제 조립 STEP Image 중심 학습

MODEL4			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	99.0	99.7	81.5
Generalization	87.8	89.3	69.0
변화	-11.0	-10.3	-12.5

MODEL5			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	98.9	99.9	99.5
Generalization	88.1	95.5	92.9
변화	-10.8	-4.4	-6.6

Model 5 Log(Recipe Test)

```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
WARNING ⚠ val: Slow image access detected (ping: 0.5±0.2 ms, read: 14.3±31.6 MB/s, size: 69.7 KB). Use local storage instead of remote/mounted
val: Scanning /content/drive/MyDrive/assembly_ai_project/Test/test/labels... 50 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 5
val: New cache created: /content/drive/MyDrive/assembly_ai_project/Test/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 4/4 1.2it/s 3.4s
                   all         50        330      0.989      0.999      0.995      0.857
           black_L_plate         30         60      0.993          1      0.995      0.808
         black_plate_1x6         20         40          1      0.986      0.995      0.745
                car_base         20         20      0.988          1      0.995      0.978
             driver_seat         30         30       0.99          1      0.995      0.861
            front_bumper         45         45          1          1      0.995      0.835
      front_windshield_roof         10         10      0.973          1      0.995      0.931
              rear_bumper         35         35      0.994          1      0.995      0.863
             rear_spoiler         15         15      0.973          1      0.995      0.834
           rear_windshield         20         20      0.988          1      0.995      0.871
      red_black_block_1x2         25         25      0.985          1      0.995      0.803
            steering_wheel         30         30       0.99          1      0.995      0.897
Speed: 4.1ms preprocess, 8.3ms inference, 0.0ms loss, 4.0ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_5/model_5_test_metrics
```

Model 5 Log(Generalization Test)

```
Ultralytics 8.4.146 🚀 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO26n summary (fused): 120 layers, 2,376,981 parameters, 0 gradients, 5.3 GFLOPs
WARNING ⚠ val: Slow image access detected (ping: 0.7±0.2 ms, read: 0.3±0.1 MB/s, size: 224.7 KB). Use local storage instead of remote/mounted
val: Scanning /content/drive/MyDrive/assembly_ai_project/Generalized Test/test/labels.cache... 18 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 7.5s/it 15.1s
                   all         18        128      0.881      0.955      0.929      0.694
           black_L_plate          8         15      0.948      0.867      0.926      0.652
         black_plate_1x6          8         16      0.743      0.812      0.886      0.354
                car_base          9          9      0.885          1      0.995      0.995
             driver_seat         15         15      0.948          1      0.995      0.847
            front_bumper         18         18       0.96          1      0.995      0.701
      front_windshield_roof          1          1      0.731          1      0.995      0.697
              rear_bumper         15         15      0.985      0.933      0.945        0.7
             rear_spoiler          3          3      0.492      0.974      0.497      0.261
           rear_windshield          5          5      0.993          1      0.995      0.822
      red_black_block_1x2         14         14          1      0.983      0.995      0.801
            steering_wheel         17         17          1      0.933      0.992        0.8
Speed: 5.4ms preprocess, 5.3ms inference, 0.0ms loss, 1.4ms postprocess per image
Results saved to /content/drive/MyDrive/assembly_ai_project/save/model_5/model_5_generalized_test_metrics
```

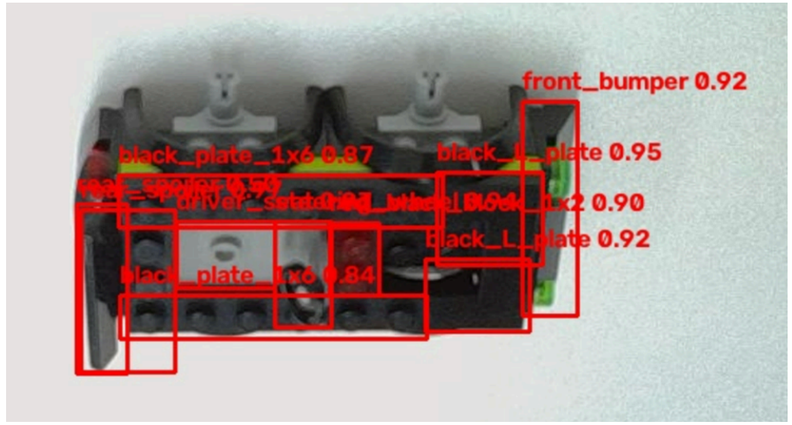
Recall +6.2 %

mAP(Recipe/Generalization) +18 % / 23.9 %

취약 부품 개선 - rear _ spoiler 오검출

취약 부품은 Targeted data 추가 → “해당 부품이 등장하는 Step 중심”으로 추가 data 수집 (전체 data X)

rear_spoiler	Generalization(%)
Precision(P)	49.2
Recall(R)	97.4
mAP50	49.7



공정 운영 → Fail Case 수집 → 취약 Class&Step 분석 → 데이터 추가 → 재학습

최종 데이터 전략

Step 중심 Dataset + Horizontal Flip

↓

신규 Recipe는 변경 Step만 추가

↓

취약 Class 발생 시 해당 Class&Step 데이터 선택적 보강

↓

운영 중 Fail Case 수집 및 재학습

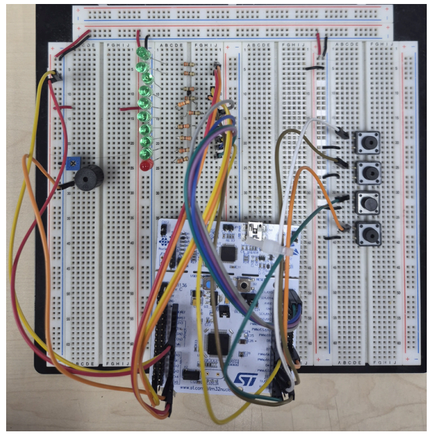
MODEL 5			
평가	Precision(%)	Recall(%)	mAP50(%)
Recipe Test	98.9	99.9	99.5
Generalization	88.1	95.5	92.9
변화	-10.8	-4.4	-6.6

최소한의 데이터 구축으로

실용적 성능 확보

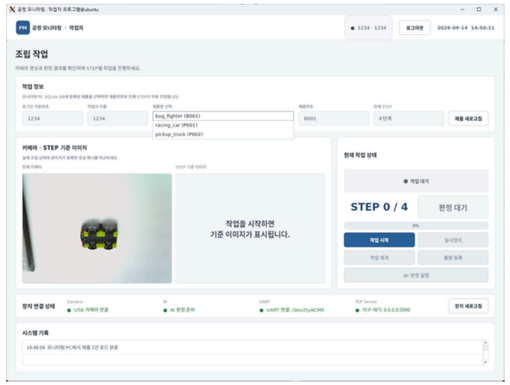
4. 시스템 기능 및 통신

시스템 기능 및 통신



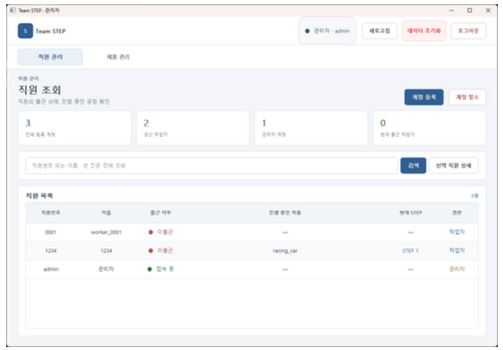
STM32

현장 제어



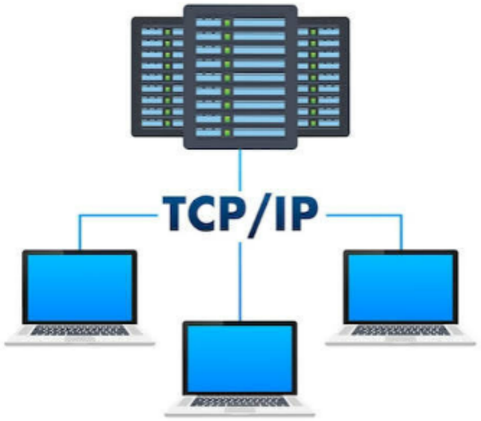
Worker PC

조립 작업을 진행하는 화면



Admin PC

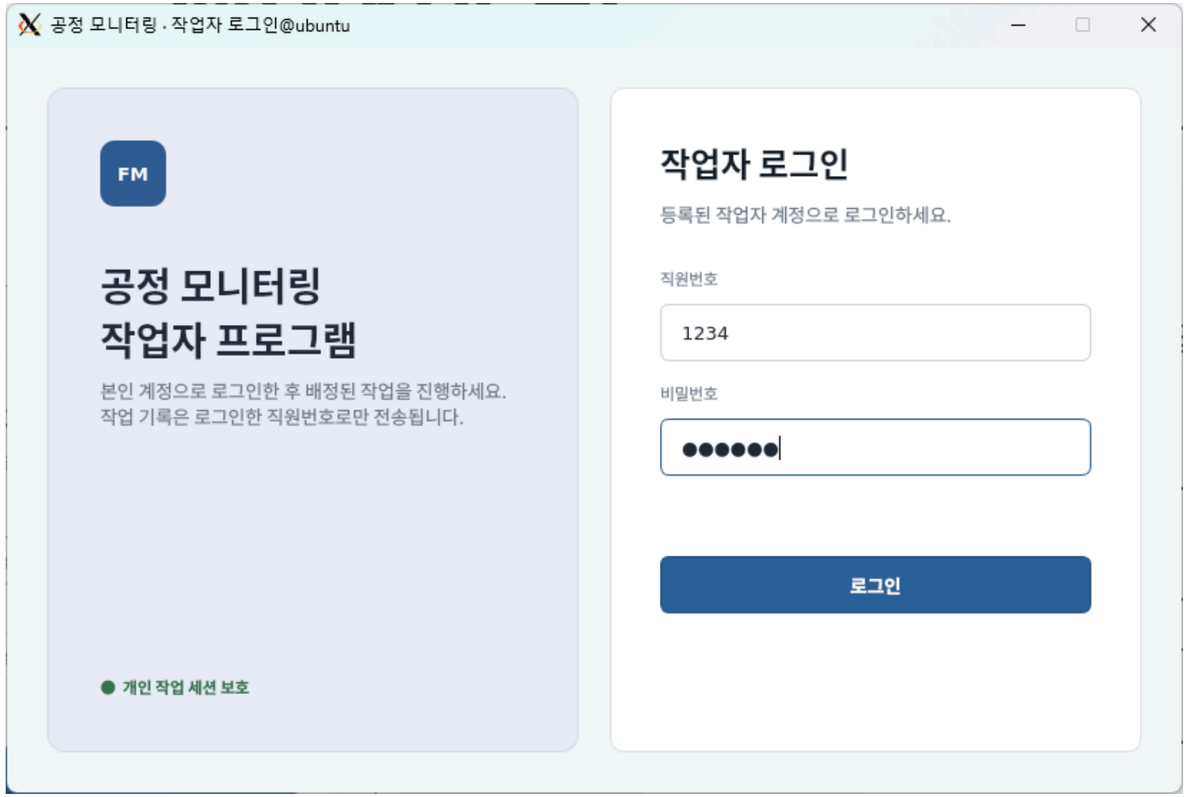
작업 이력과 제품 품질 관리



통신 프로토콜

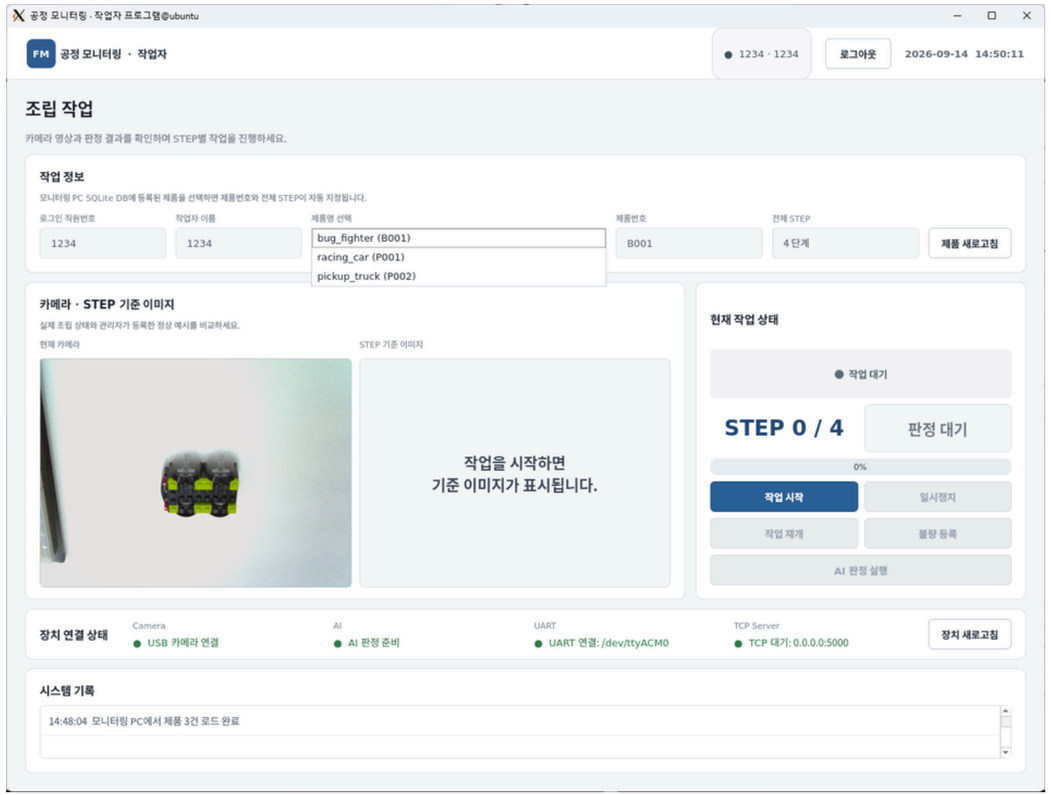
- UART 패킷: STM32 ←→ Worker PC
- TCP/IP 패킷: Worker PC ←→ Admin PC

Worker PC



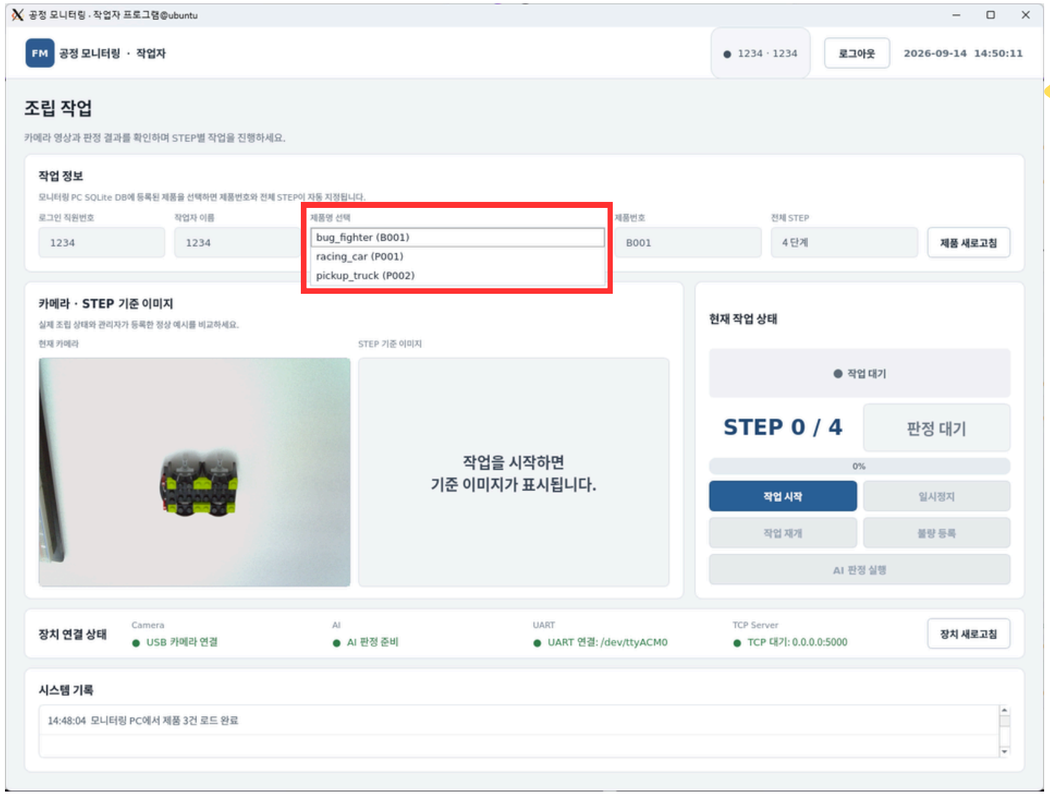
로그인 페이지

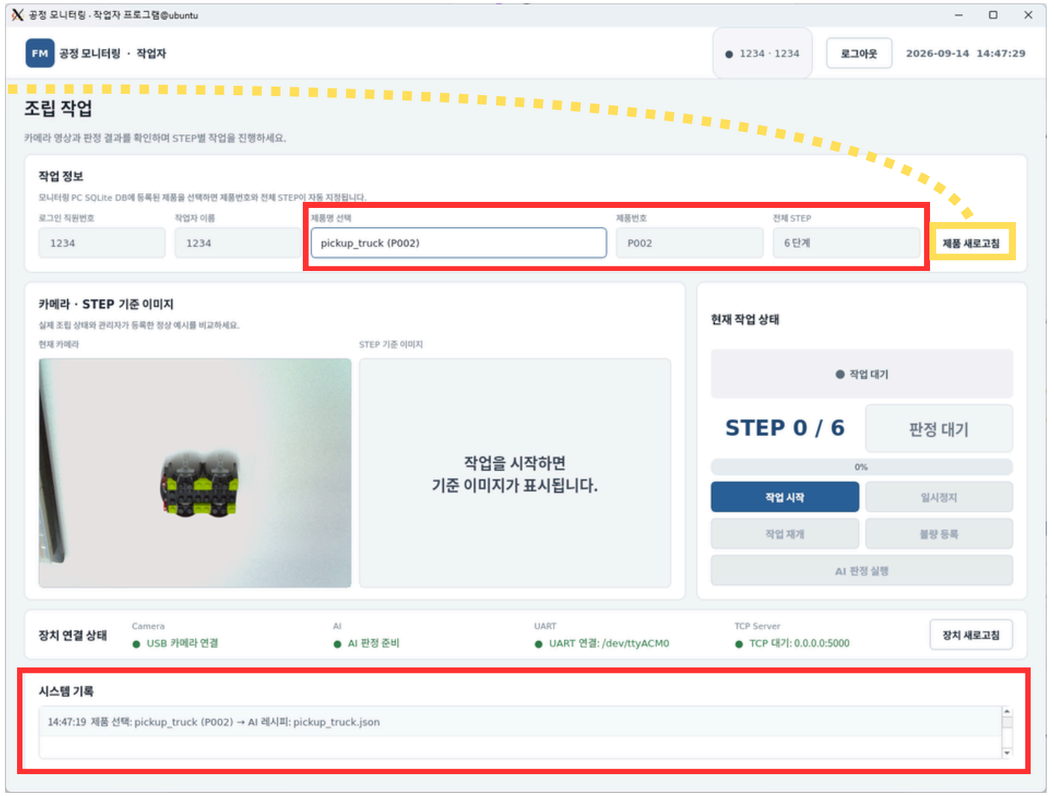
Worker PC



Worker 기본 화면

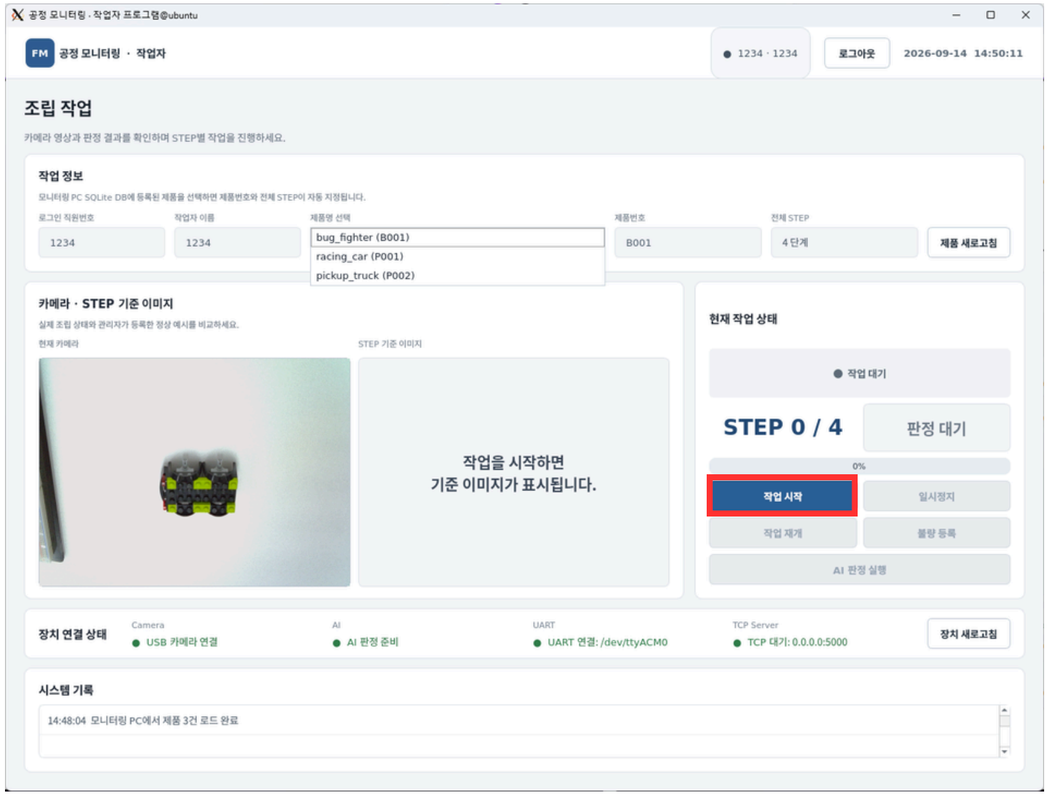
Worker PC

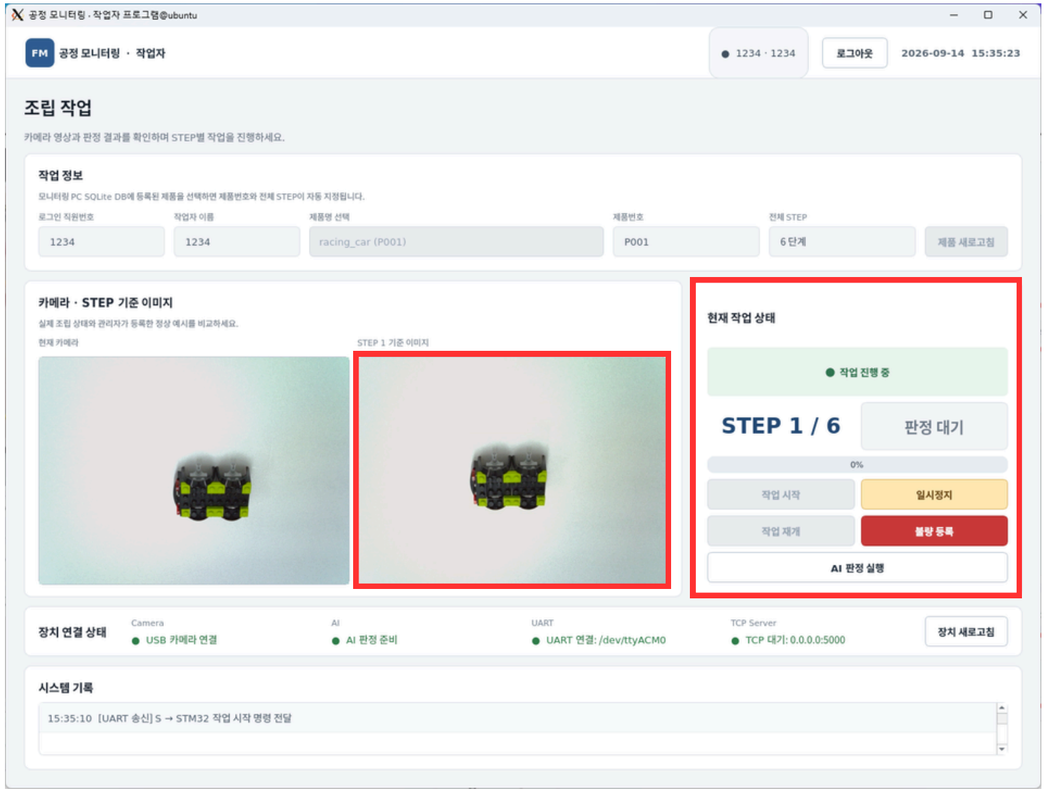




제품 선택

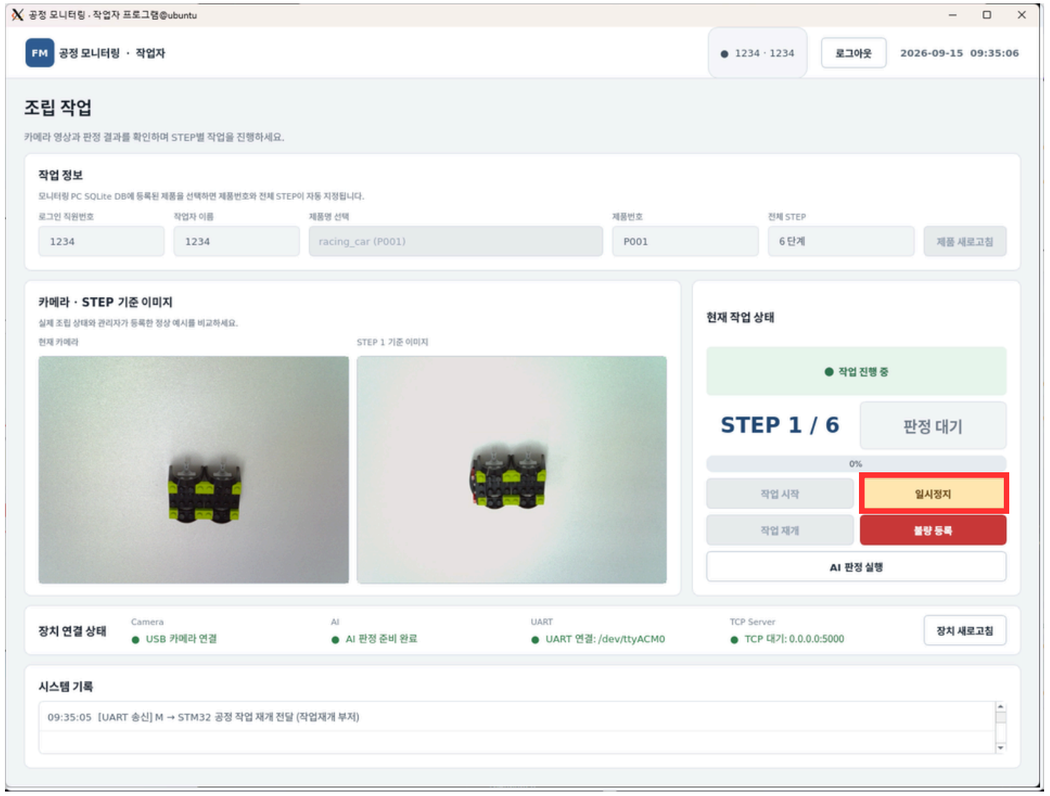
Worker PC



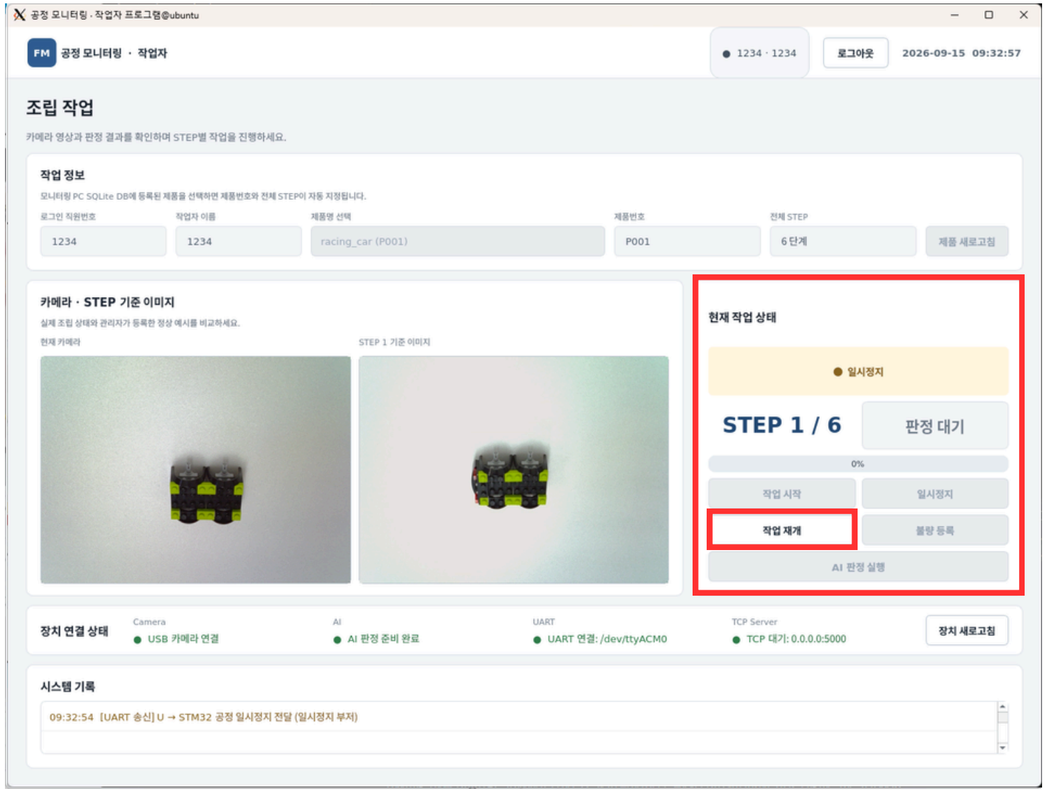


작업 시작

Worker PC

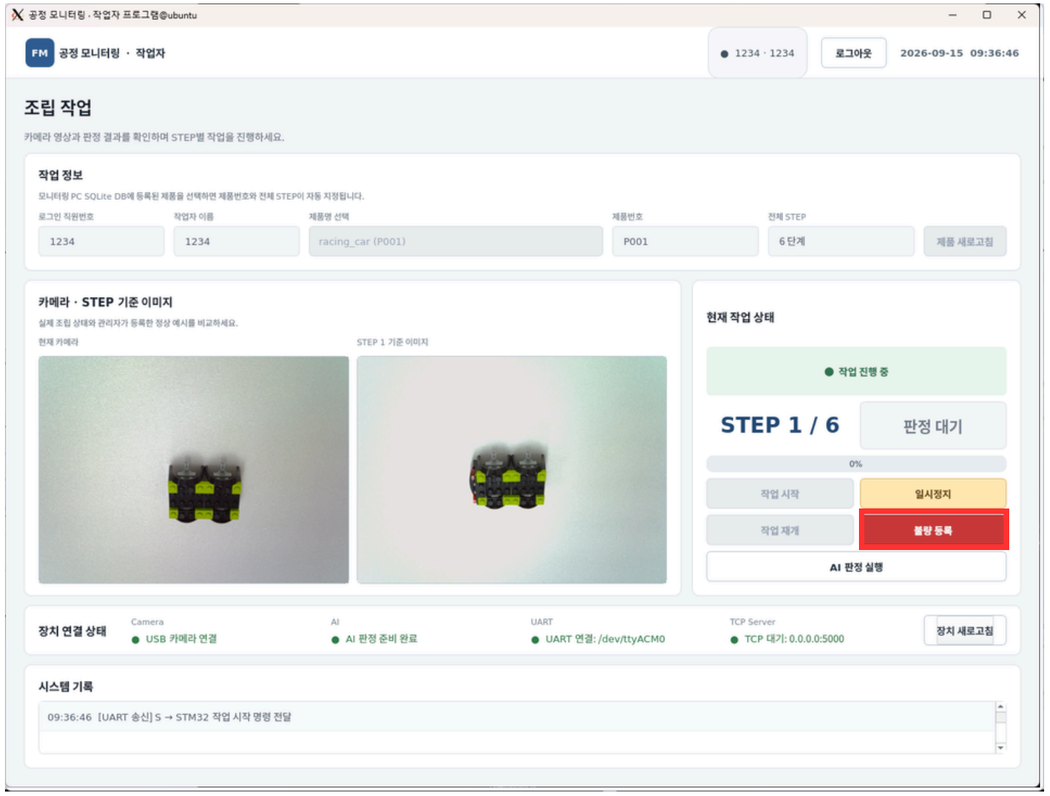


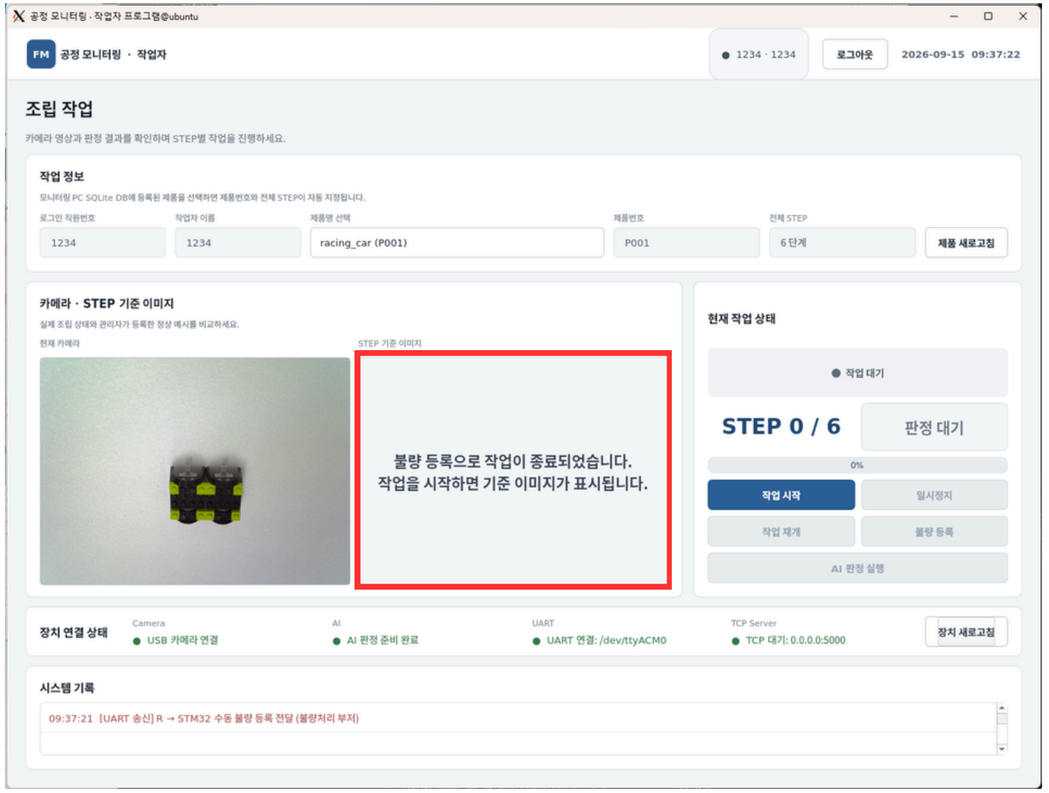
일시정지



작업 재개

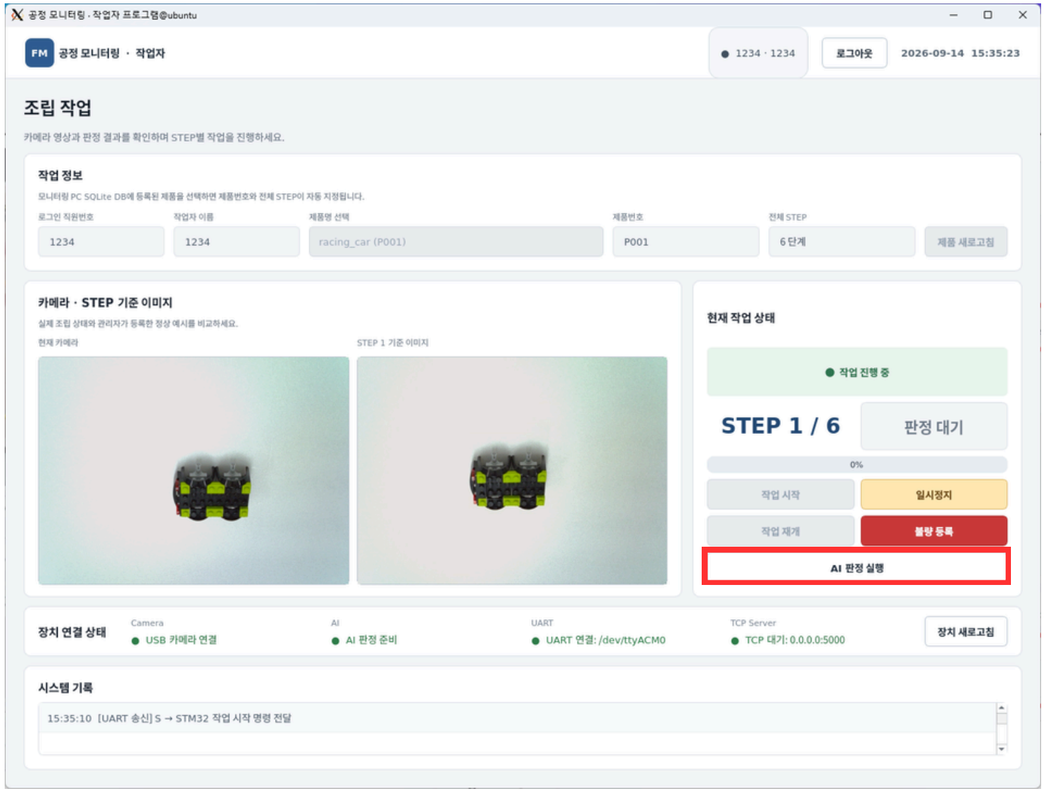
Worker PC

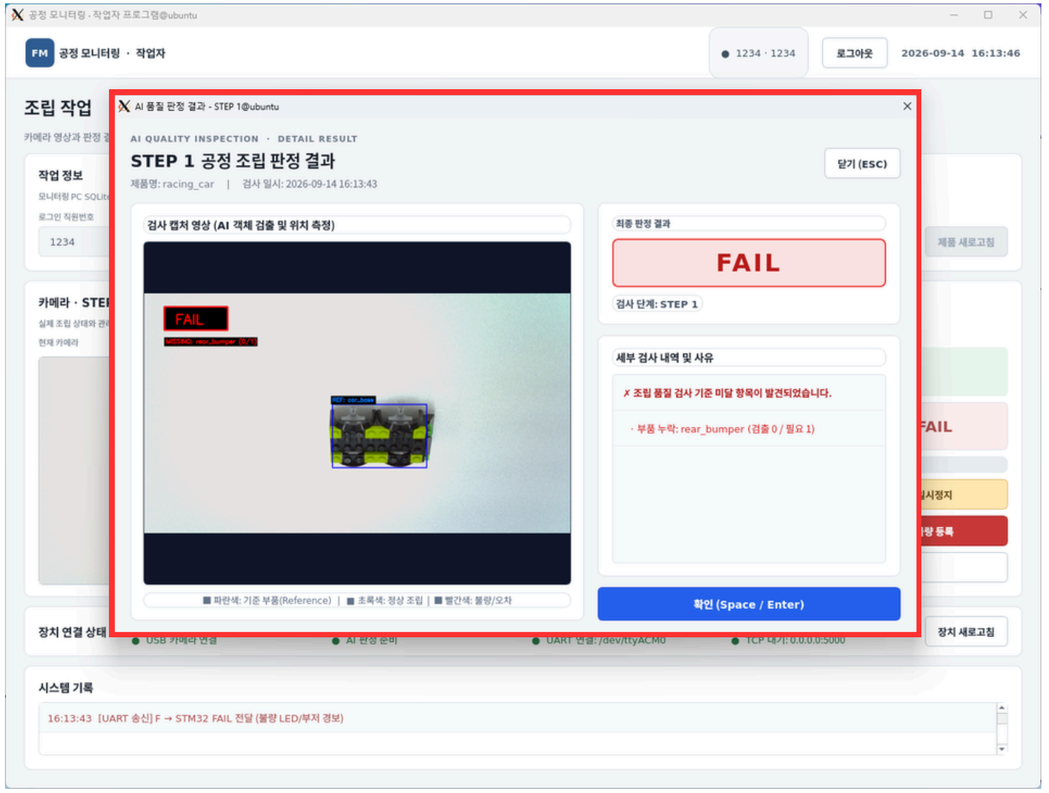




불량 등록

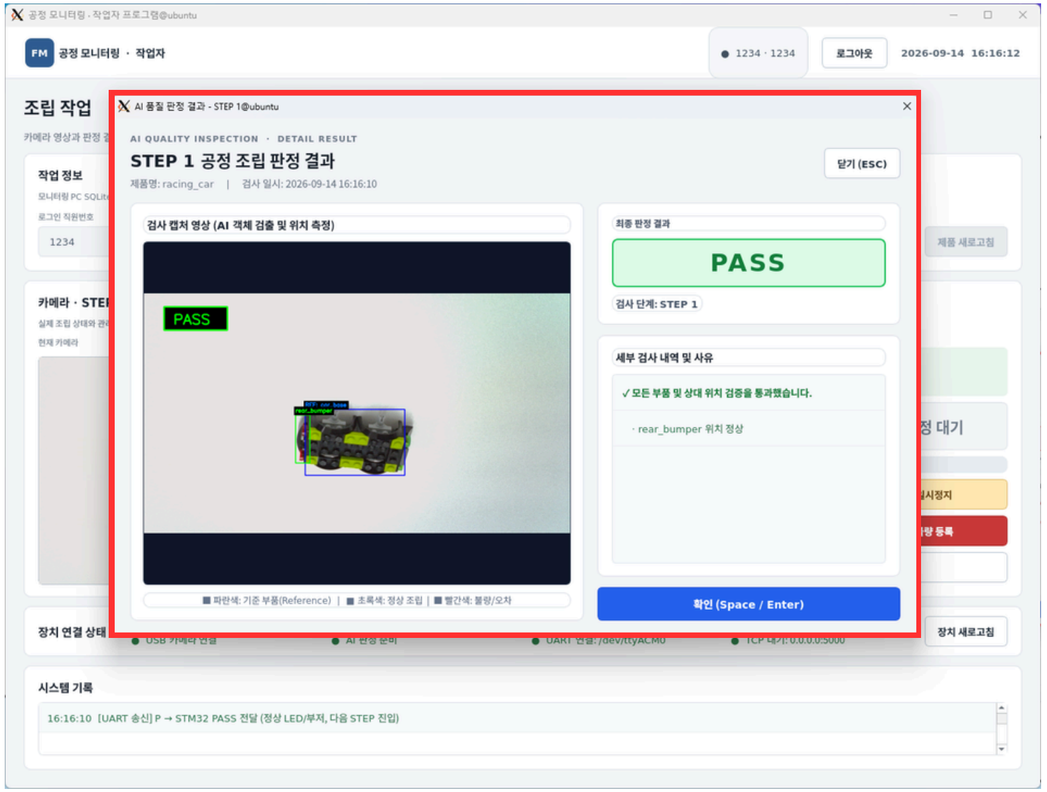
Worker PC

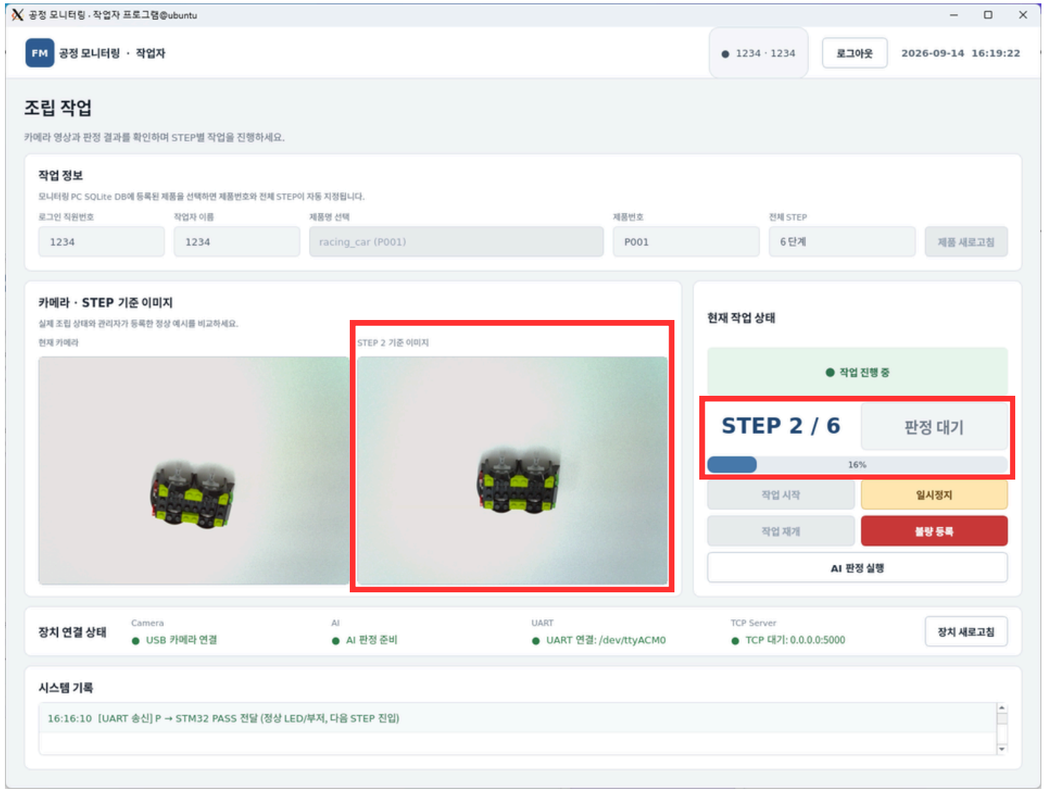




PASS/FAIL Check
(FAIL)

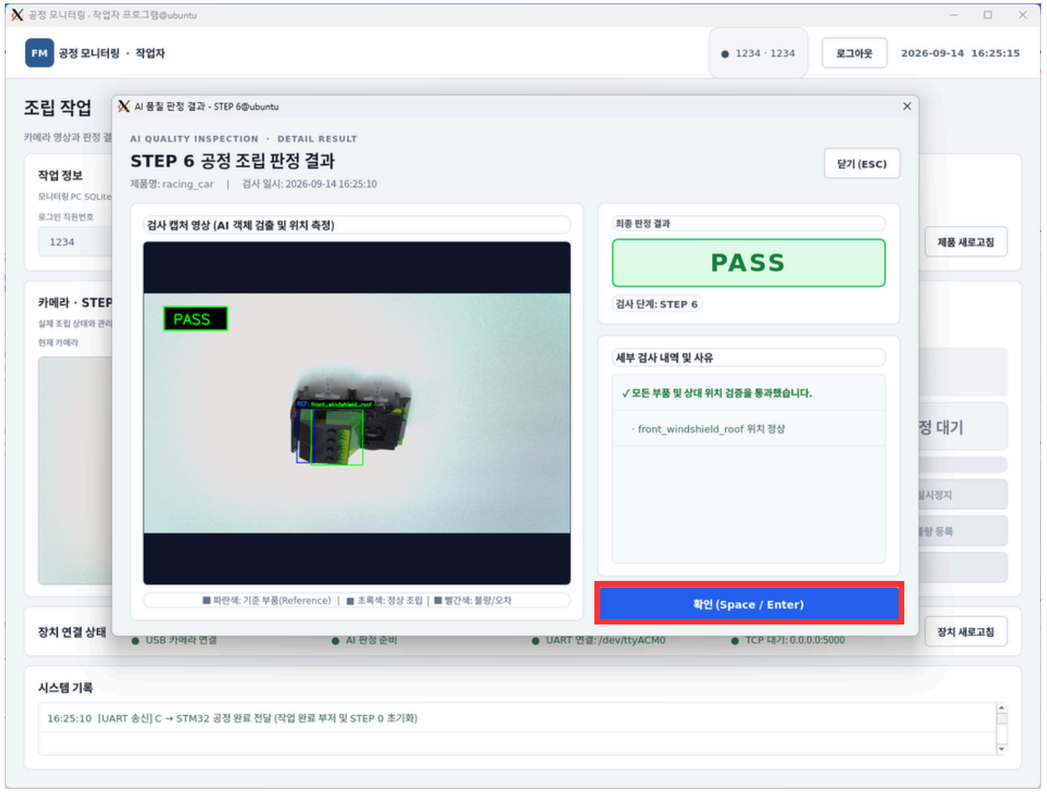
Worker PC

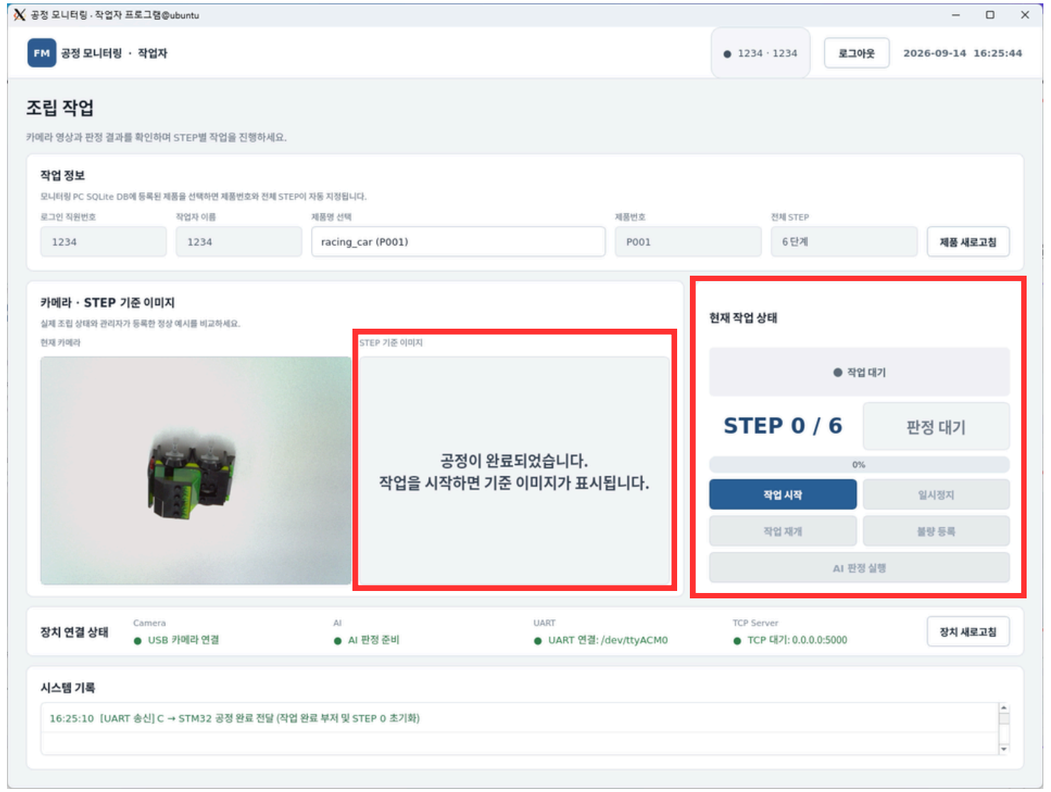




PASS/FAIL Check
(PASS)

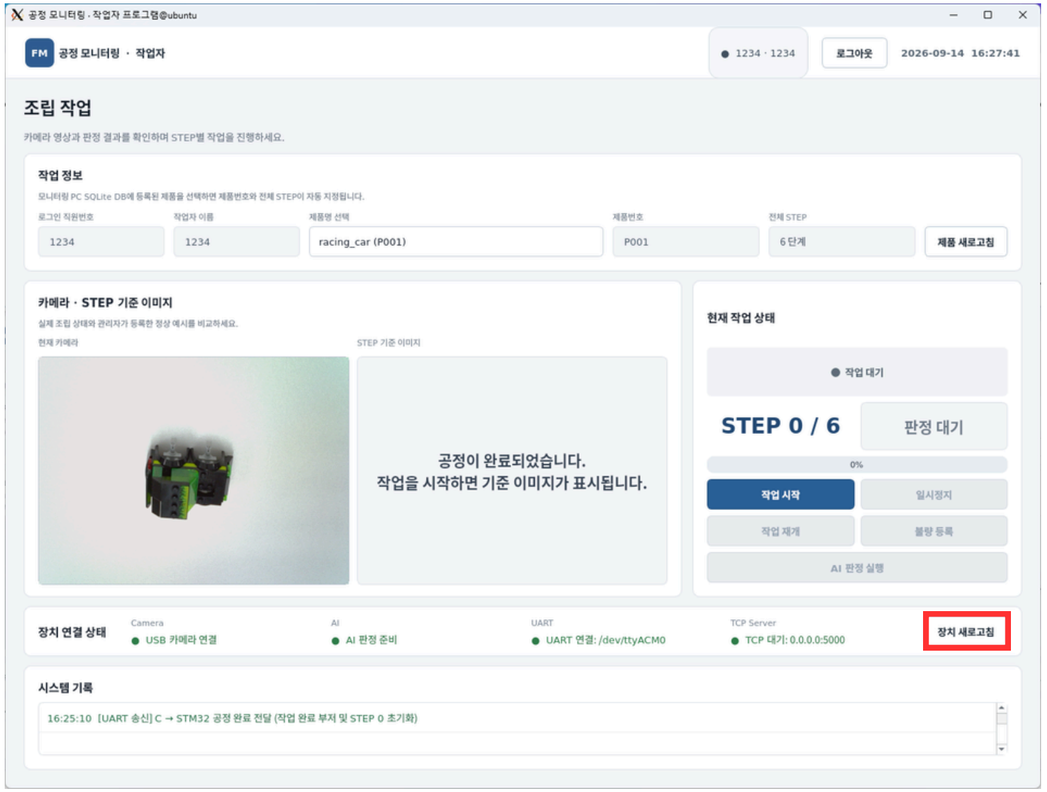
Worker PC

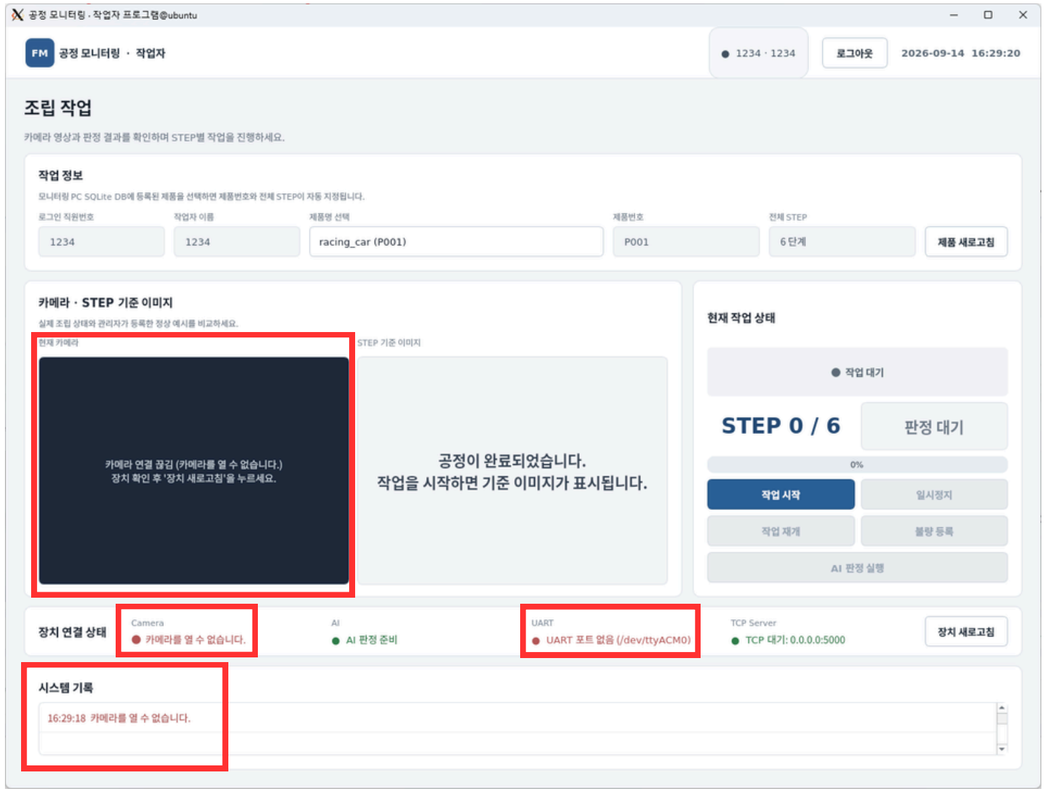




PASS/FAIL Check
(COMPLETE)

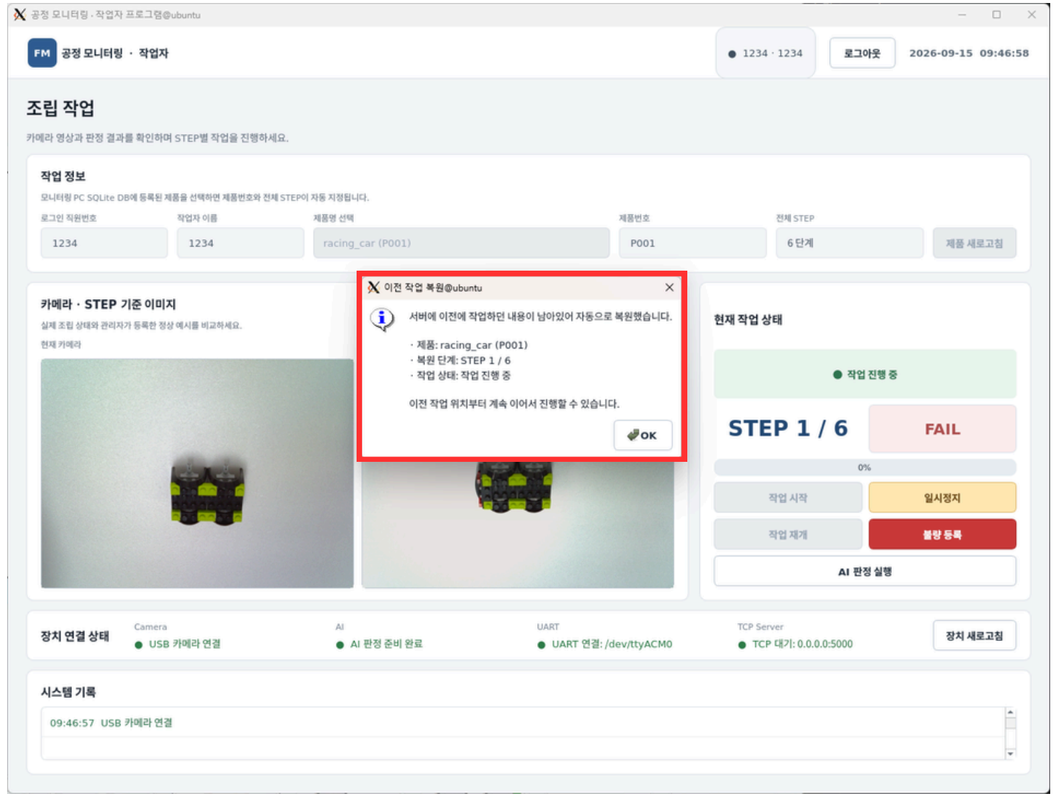
Worker PC





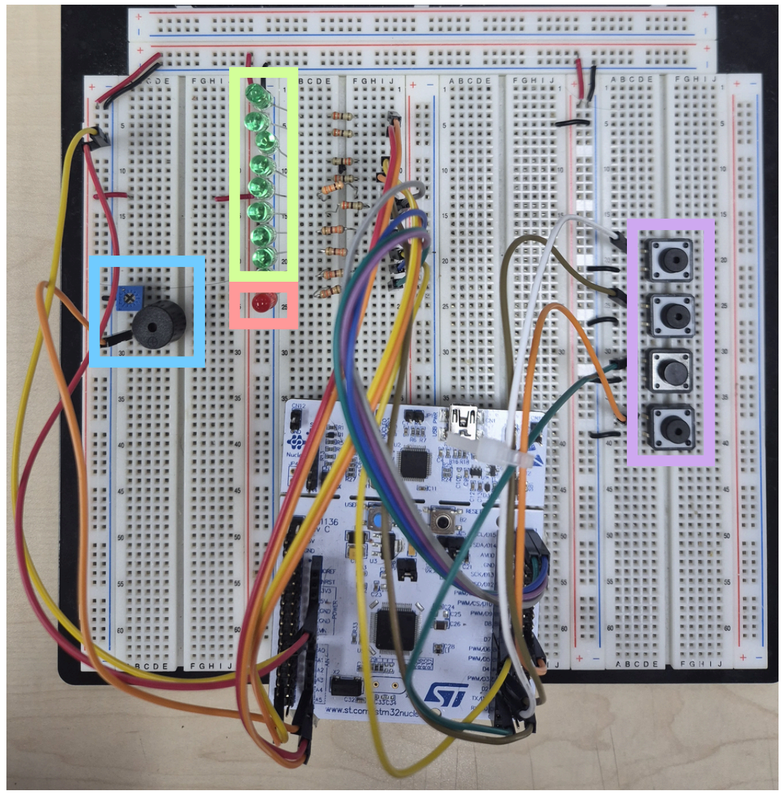
장치 연결 상태

Worker PC



작업 기록 복구

STM32



LED _ STEPx(x = 1 ~ 9)

LED _ FAIL

Buzzer & Variable Resistor

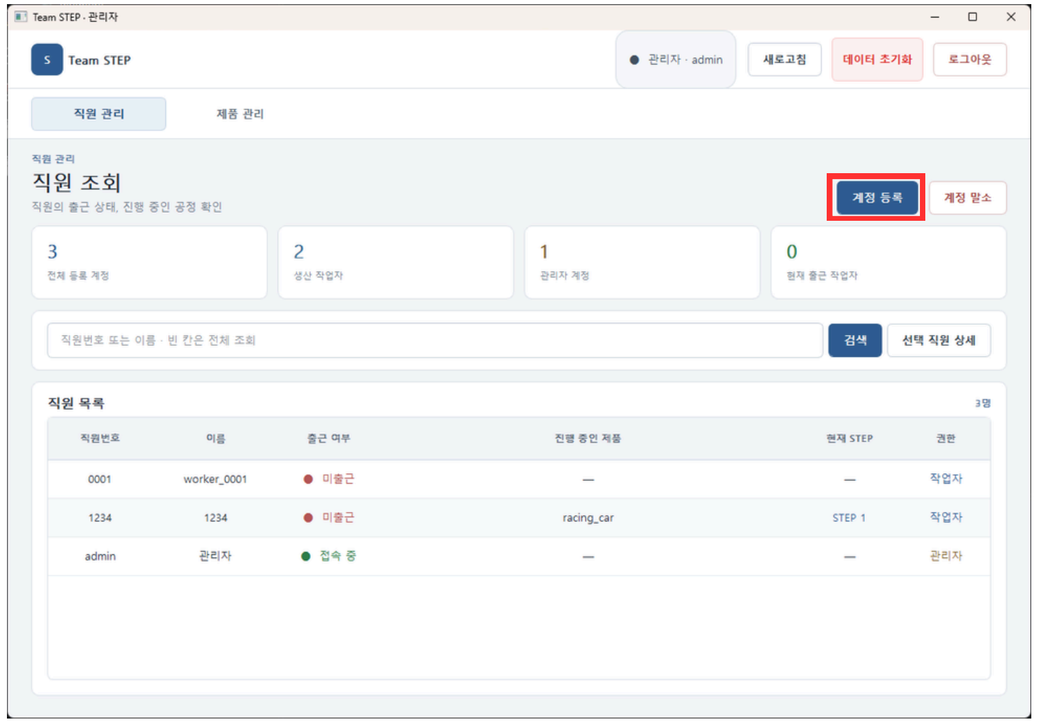
Button _ Group

STM 모듈 설명

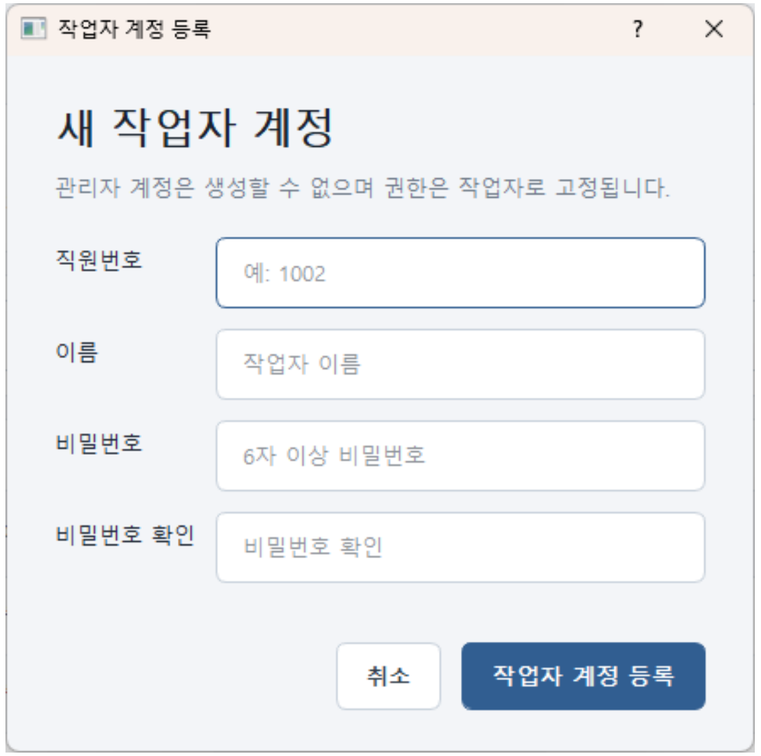
Monitoring PC

로그인 페이지

Monitoring PC

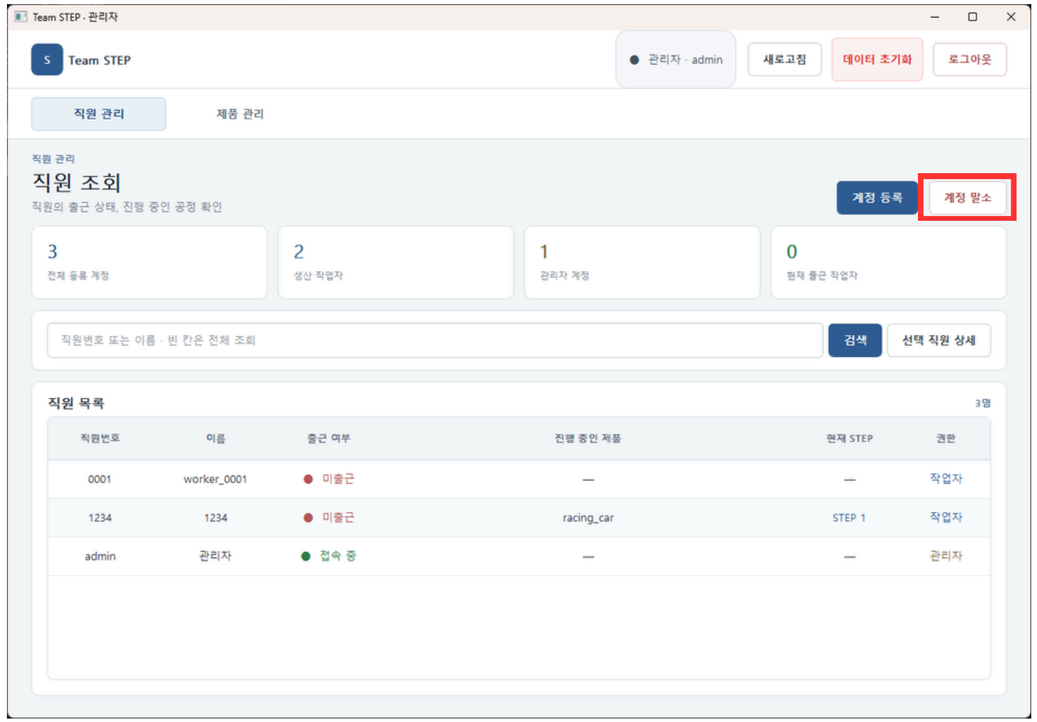


직원 관리

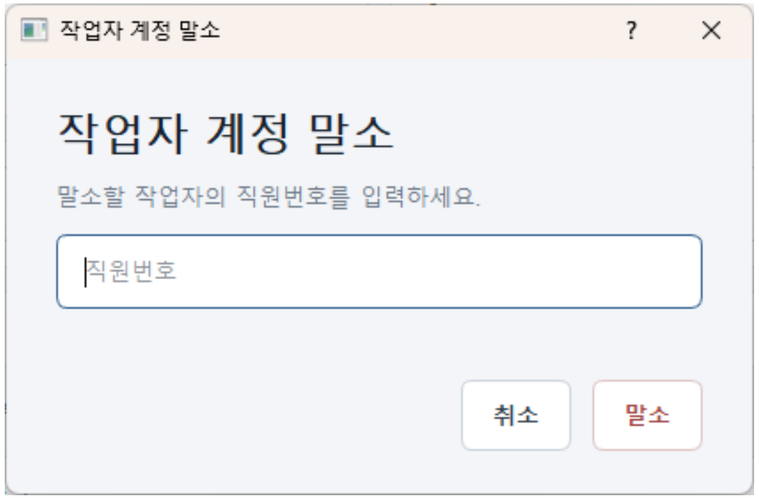


작업자 등록

Monitoring PC

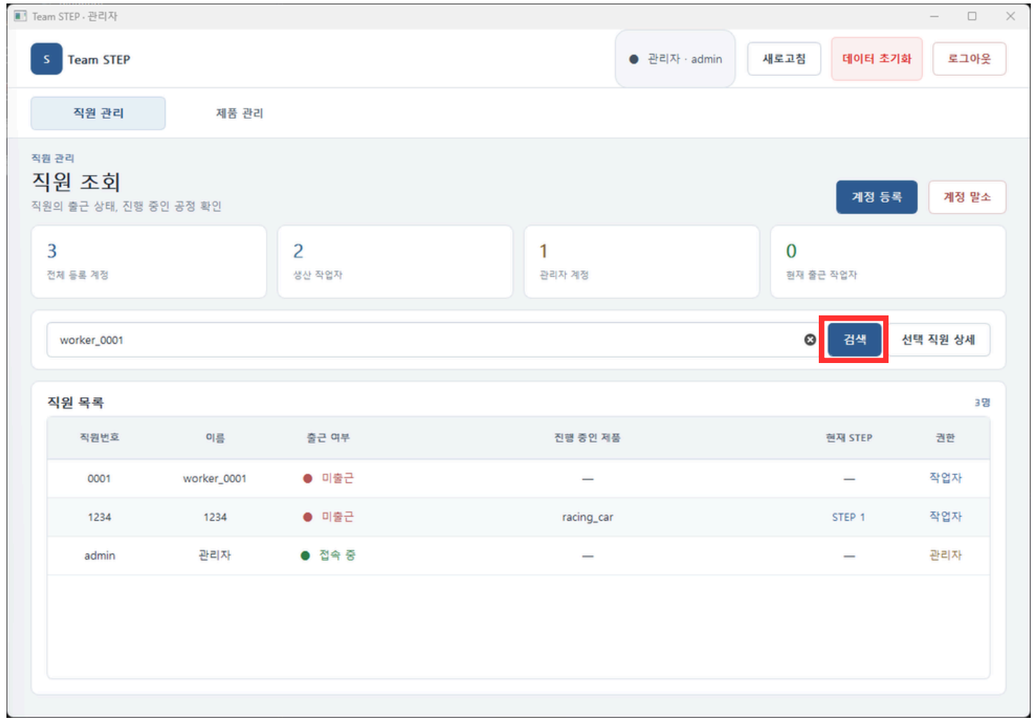


직원 관리

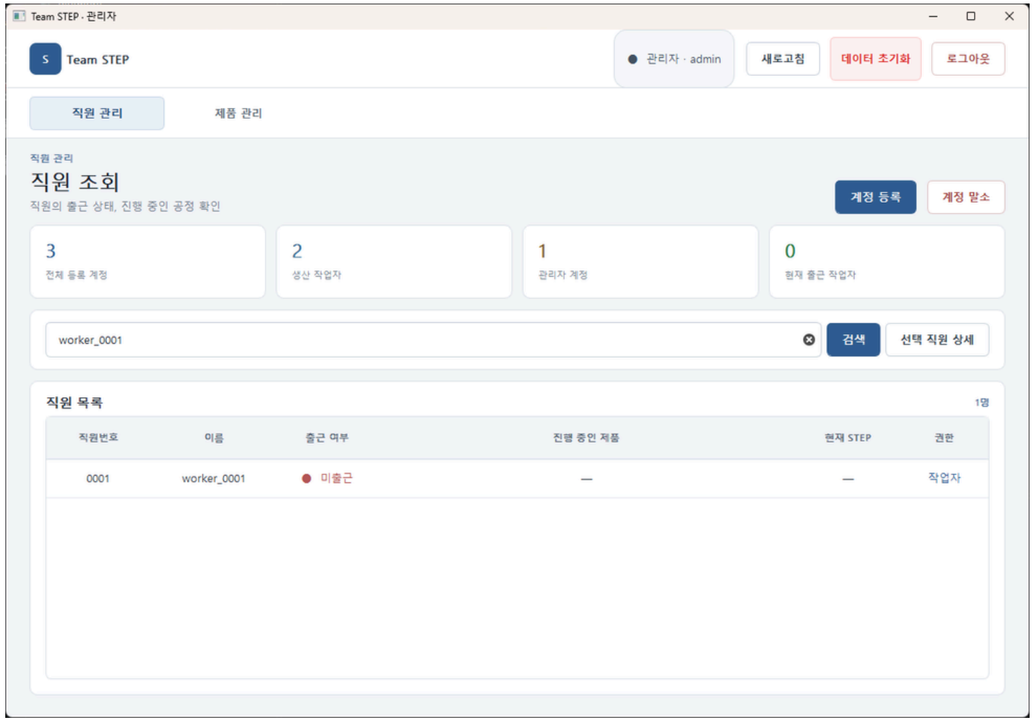


작업자 말소

Monitoring PC

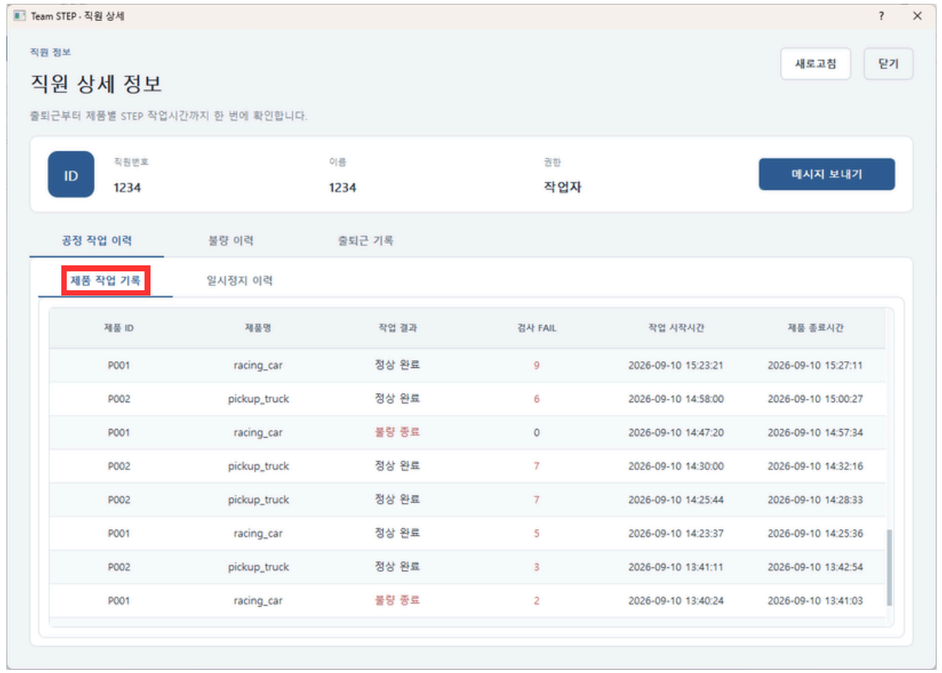


직원 조회

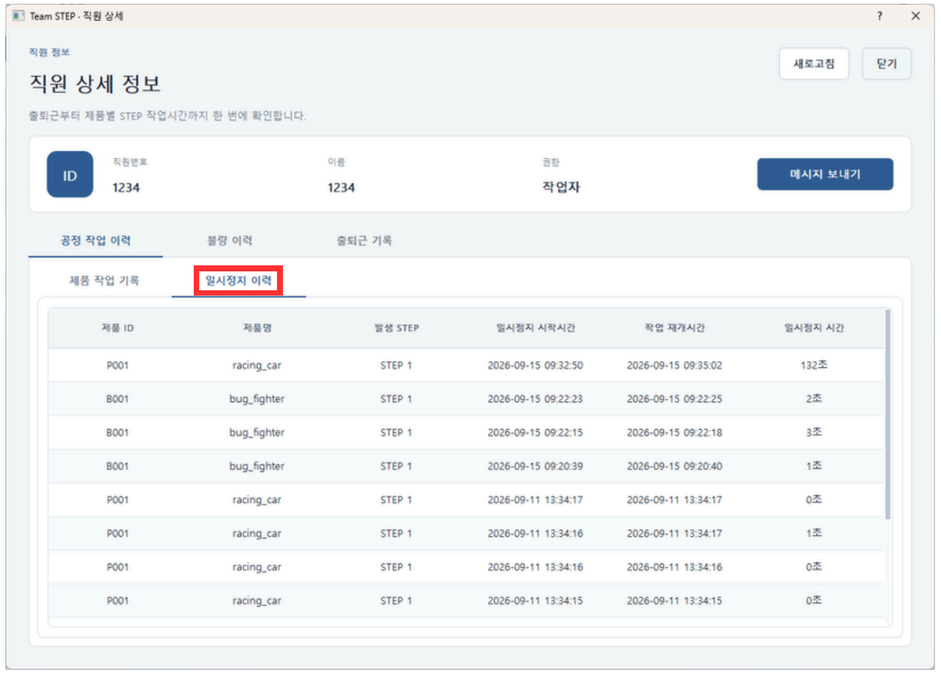


직원 검색

Monitoring PC

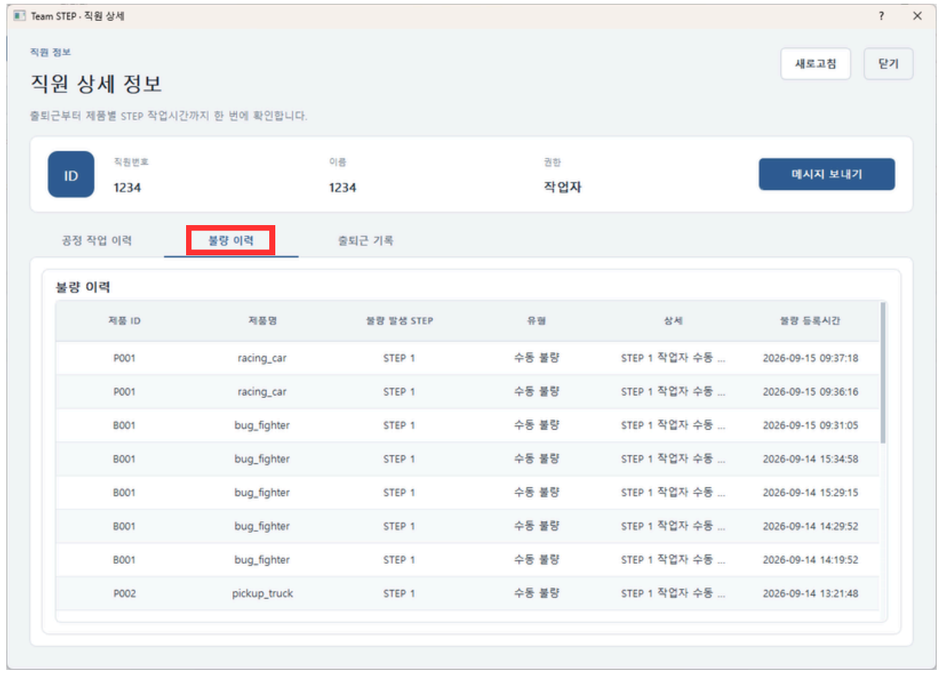


직원 상세 정보
(제품 작업 기록)

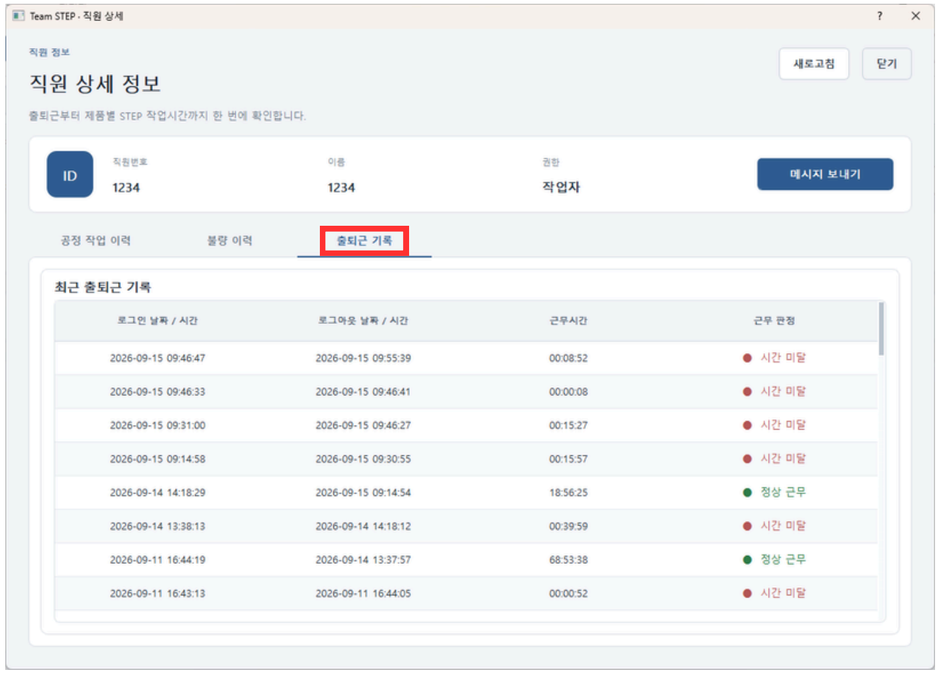


직원 상세 정보
(일시정지 이력)

Monitoring PC

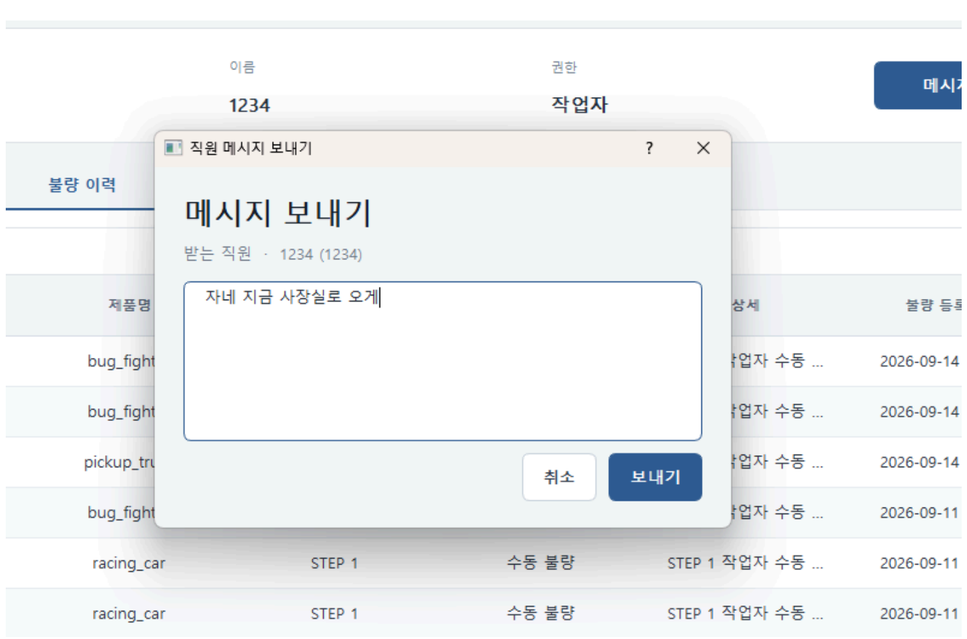


직원 상세 정보
(불량 이력)

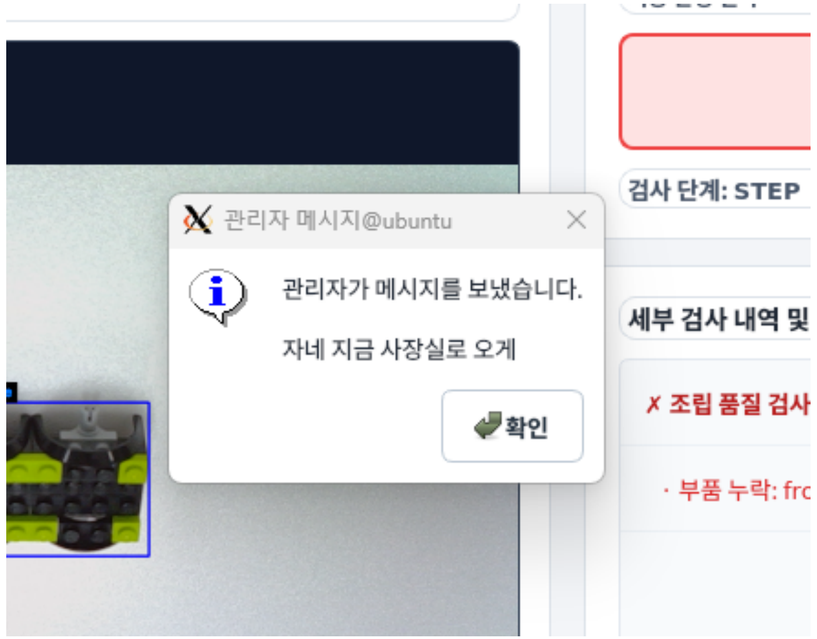


직원 상세 정보
(출퇴근 기록)

Monitoring PC

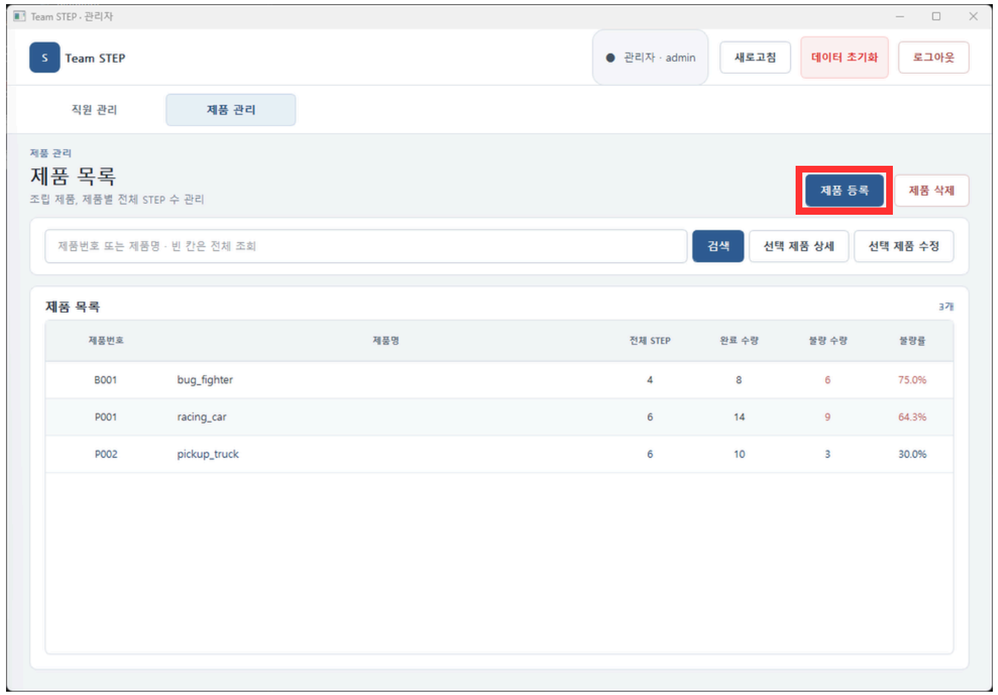


작업자 호출
(관리자 화면)



작업자 호출
(작업자 화면)

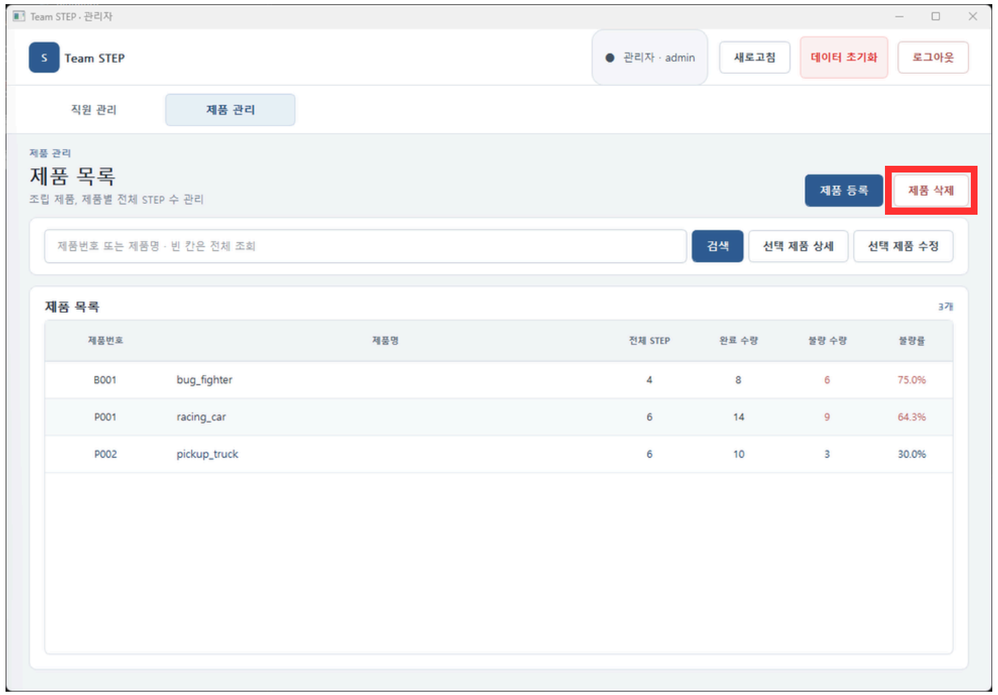
Monitoring PC



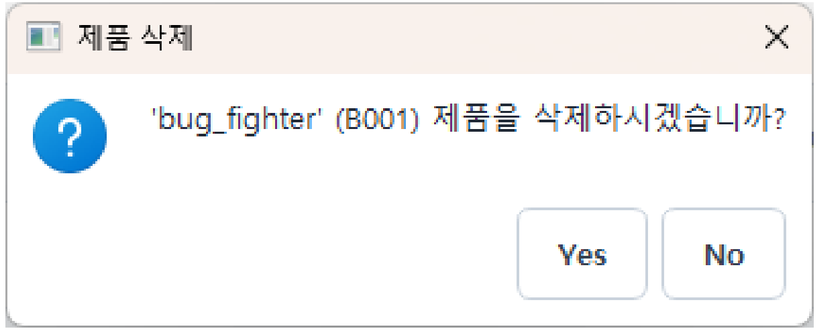
제품 관리 페이지

제품 등록

Monitoring PC

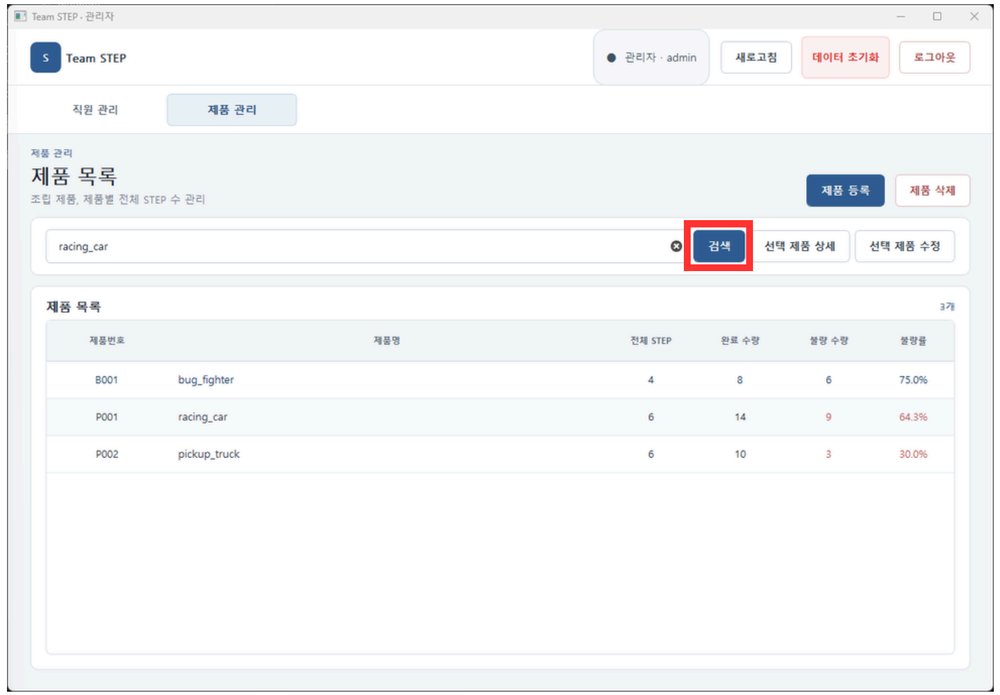


제품 관리 페이지

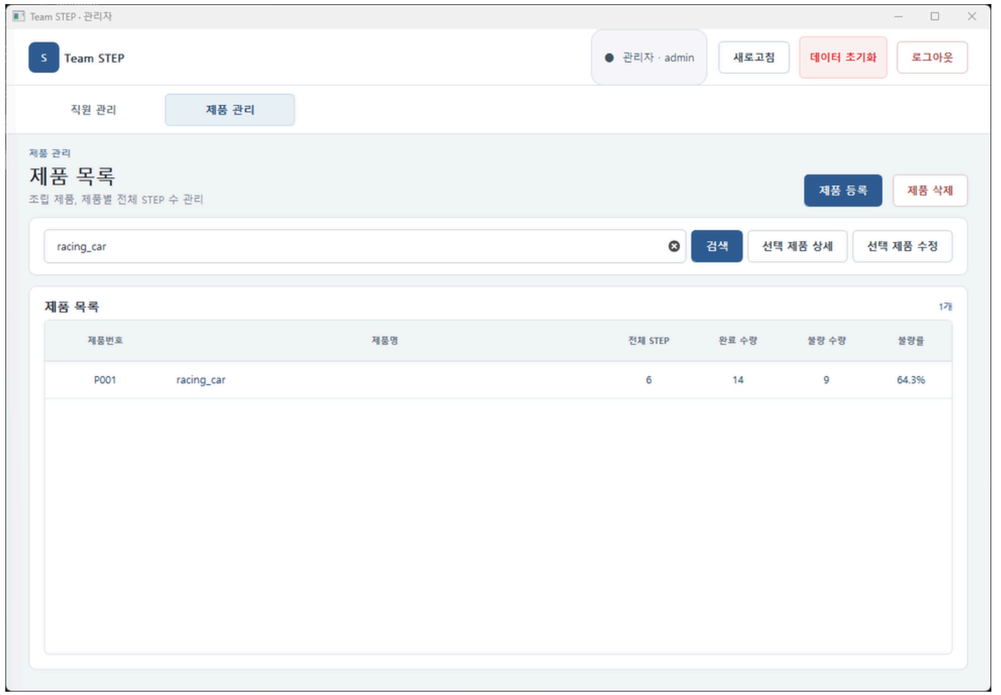


제품 삭제

Monitoring PC

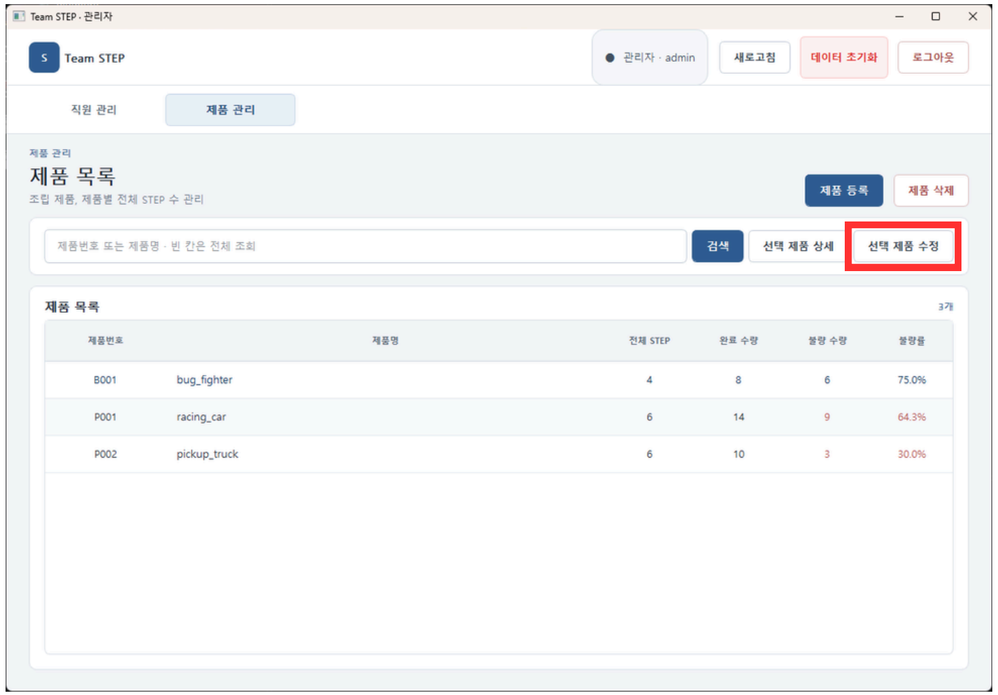


제품 조회



제품 검색

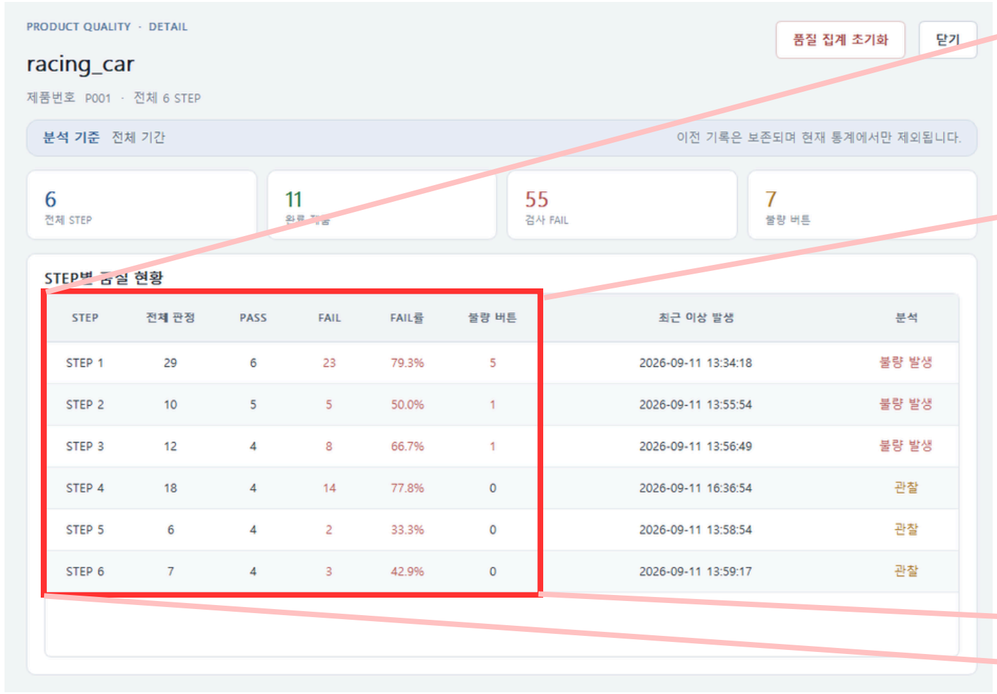
Monitoring PC



제품 수정

제품 수정 팝업

Monitoring PC



제품 상세 정보

STEP	전체 판정	PASS	FAIL	FAIL률	불량 버튼
STEP 1	29	6	23	79.3%	5
STEP 2	10	5	5	50.0%	1
STEP 3	12	4	8	66.7%	1
STEP 4	18	4	14	77.8%	0
STEP 5	6	4	2	33.3%	0
STEP 6	7	4	3	42.9%	0

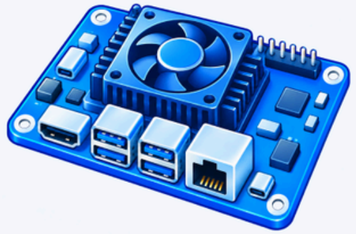
스텝 별 Fail, 불량률

TCP / IP 통신

Worker (Jetson)
제품 확인, 작업 진행
5001 작업 요청 → 응답
로그인
아이디, 비밀번호
인증결과, 이전 작업 정보
작업 준비
제품, STEP 이미지 요청
제품 목록, 기준 이미지
작업 상태 전송
현재 STEP, 작업 상태, 검사결과
처리 결과
5000 관리자 메세지
메세지 전달
관리자 메세지를 작업자 화면에 전달
Admin
작업 현황 확인, 관리
DB
작업자 계정, 제품, STEP 정보
근무 시간, 작업 진행 이력
검사 결과, 불량 이력
5001 : 관리자가 수신
5000 : 작업자가 수신

UART 통신

Jetson



AI 검사, STEP 관리
작업 상태 처리

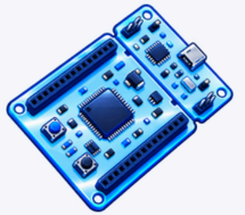
버튼 입력

Start : 작업 시작 요청
Check : AI 검사 요청
Pause / Resume : 정지, 재개 요청
Reset : 불량 처리 요청

결과, 제어 명령

S : 시작, P : 다음 STEP, F : FAIL
C : 완료, R : 불량
U / M : 정지 / 재개, E : 종료
1~9 : STEP 복원

STM32



버튼 LED 부저

버튼 입력 감지
STEP LED, FAIL LED, 부저 제어

검사 버튼 → Check 전송 → Jetson AI 검사 → PASS / FAIL 전송 → LED, 부저 표시

5. 결론

기대 효과

조립 품질 향상	부품 누락·오조립을 공정 중에 발견해 불량품 발생 및 재작업률 감소	>
작업 효율 및 표준화	STEP별 안내와 검사로 작업자 숙련도 차이 보완 및 일관된 조립 절차 유지	>
현장 관리 효율 향상	작업자의 진행·정지 상태를 실시간으로 확인하여 지연이나 이상 상황에 빠르게 대응	>
데이터 기반 공정 개선	작업 시간과 검사·불량 이력을 분석하여 반복되는 문제와 병목 구간 파악 및 개선	>

느낀 점

고지현

- 무조건 성능이 좋은 AI 모델을 쓰는 것이 아닌 용도에 맞게 적절한 모델을 고르는 것이 연산 속도나 모델의 용량 면에서 더 이점을 얻을 수 있다는 것을 느낌
- Stm32 코딩에서 딜레이를 Timer Interrupt로 처리할 때는 Button Debouncing이나 Interlock, flag 변수 등을 잘 설정해야 한다는 것을 깨달음

김선호

- 모니터링 화면과 TCP/IP 통신, DB를 구현하면서 작업자의 상태가 관리자 화면에 전달되는 과정을 이해할 수 있었음
- 데이터가 중복 집계되는 문제를 해결하며, 통신은 데이터를 보내는 것뿐 아니라 받은 데이터를 정확하게 처리하는 것도 중요하단 것을 배움

박상현

- 카메라 스트리밍과 딥러닝 추론, 통신이 얽히며 발생한 UI 프리징을 멀티스레딩(QThread) 아키텍처로 분리해 해결
- 더불어 조도 환경 변화로 인한 인식 오류를 리눅스 v4l2-ctl로 하드웨어 단에서 직접 보정하면서, 단순 애플리케이션 코딩을 넘어 OS와 디바이스를 아우르는 시스템 엔지니어링 시야를 기름

윤홍은

- AI 모델의 성능뿐만 아니라 데이터 구성, 촬영 환경, 판정 로직 등 시스템 전체를 함께 고려하는 것이 중요하다는 점을 배움

Thank You

감사합니다